



Review

Fate and transport modelling for evaluating antibiotic resistance in aquatic environments: Current knowledge and research priorities

Mahesh Jampani^{a,*}, Javier Mateo-Sagasta^a, Aparna Chandrasekar^{b,c}, Despo Fatta-Kassinos^d, David W. Graham^e, Ritu Gothwal^a, Arshnee Moodley^{f,g}, Vishnumurthy Mohan Chadag^h, David Wiberg^a, Simon Langan^a

^a International Water Management Institute (IWMI), Battaramulla, Colombo, Sri Lanka

^b UFZ - Helmholtz Centre for Environmental Research, Department Computational Hydrosystems, Leipzig, Germany

^c Institute of Hydrobiology, Technische Universität Dresden, Dresden, Germany

^d Civil and Environmental Engineering Department and Nireas International Water Research Center, University of Cyprus, Nicosia, Cyprus

^e School of Engineering, Newcastle University, Newcastle Upon Tyne, United Kingdom

^f International Livestock Research Institute (ILRI), Nairobi, Kenya

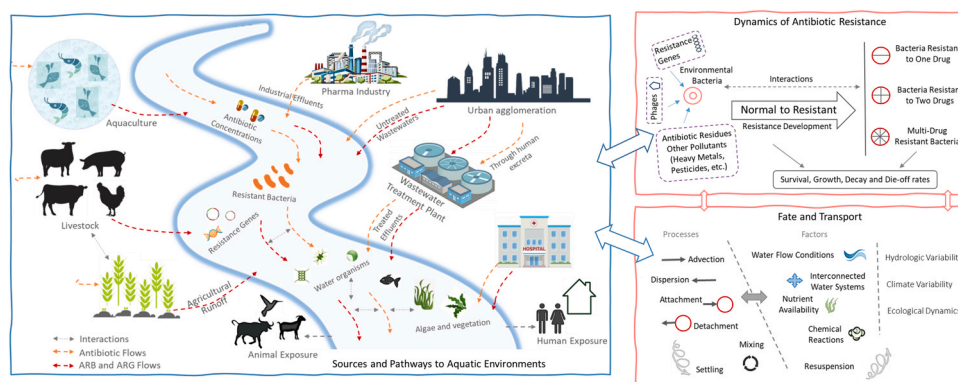
^g Department of Veterinary and Animal Sciences, University of Copenhagen, Frederiksberg C, Denmark

^h WorldFish, Jalan Batu Maung, Batu Maung, Bayan Lepas, Penang, Malaysia

HIGHLIGHTS

- Evaluating the fate and transport of ARB and ARGs in water environments is critical to mitigating human health risks.
- Presented different environmental matrices that contribute to antibiotic resistant pollution.
- Discussed the critical role of environmental factors in the persistence, survival, and decay of ARB in water bodies.
- Discussed the current knowledge of AR modelling, complexities of resistance mechanisms, and limitations to overcome.
- Efficient water quality model development is vital to assess the emergence and spread of AR.

GRAPHICAL ABSTRACT



ARTICLE INFO

Editor: Jianhua Guo

Keywords:

Antibiotics
Antibiotic resistant bacteria (ARB)
Antibiotic resistance genes (ARGs)
Resistance evolution mechanisms

ABSTRACT

Antibiotics have revolutionised medicine in the last century and enabled the prevention of bacterial infections that were previously deemed untreatable. However, in parallel, bacteria have increasingly developed resistance to antibiotics through various mechanisms. When resistant bacteria find their way into terrestrial and aquatic environments, animal and human exposures increase, e.g., via polluted soil, food, and water, and health risks multiply. Understanding the fate and transport of antibiotic resistant bacteria (ARB) and the transfer mechanisms of antibiotic resistance genes (ARGs) in aquatic environments is critical for evaluating and mitigating the risks of resistant-induced infections. The conceptual understanding of sources and pathways of antibiotics, ARB, and

* Correspondence to: International Water Management Institute (IWMI), Colombo, Sri Lanka.

E-mail address: m.jampani@cgiar.org (M. Jampani).

<https://doi.org/10.1016/j.jhazmat.2023.132527>

Received 21 November 2022; Received in revised form 3 August 2023; Accepted 9 September 2023

Available online 11 September 2023

0304-3894/© 2023 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

Gene transfer
Water quality modelling

ARGs from society to the water environments is essential for setting the scene and developing an appropriate framework for modelling. Various factors and processes associated with hydrology, ecology, and climate change can significantly affect the fate and transport of ARB and ARGs in natural environments. This article reviews current knowledge, research gaps, and priorities for developing water quality models to assess the fate and transport of ARB and ARGs. The paper also provides inputs on future research needs, especially the need for new predictive models to guide risk assessment on AR transmission and spread in aquatic environments.

1. Introduction

The discovery of antibiotics revolutionised human medicine and enabled the treatment and prevention of previously untreatable microbial infections [1]. Antibiotics are extensively used in animals/livestock farming to promote growth and prevent diseases [2,3]. Many antibiotics have become ineffective against various infections that were previously easily treatable, thus warranting a need for constant renewal and update. Yet, few new antibiotics have come to the market for treating infectious diseases in recent years [4]. This has raised concerns about entering a new post-antibiotic era, where the infections caused by resistant strains are untreatable [5-7].

Recent estimates suggest that about 1.27 million additional deaths occurred in 2019 due to antibiotic-resistant infections [8], which is projected to increase to over 10 million by 2050 [7]. With the increase in antibiotic resistance (AR), hospitalisation costs also tend to increase, significantly impacting the world (up to 4% of global GDP by 2050) and individual country economies [9]. This is particularly costly to economies in the emerging and developing world, which often have large workforces and inadequate public health care provision [10]. AR development is a natural evolutionary process, but imprudent antibiotic use and wider pollution have accelerated AR emergence, including wholly new resistance in many human, animal, and plant pathogens [11-13].

Increases in antibiotic resistant bacteria (ARB) and antibiotic resistance genes (ARGs) are a clear product of global dependency on antibiotics, including almost indestructible ‘superbugs’, which are bacteria that exhibit resistance to multiple antibiotics [14]. Bacteria develop resistance through evolution (mutation) and by acquiring resistance through gene transfer. AR is acquired via vertical (e.g., from mother to daughter cells) and horizontal transfer of ARGs from other bacterial cells in close proximity [15]. Example mechanisms of horizontal gene transfer (HGT) are conjugation, transformation, and transduction (Table 1). In HGT, ARGs are often carried on mobile genetic elements

(MGEs) that can act as vectors for gene transfer to related microbial species and other strains or genera [16,17].

The natural environment plays a critical role in developing and spreading AR by being a reservoir of ARGs [25,26]. ARGs can be acquired by non-pathogenic bacteria and then be transferred to clinically relevant pathogens that can maintain and spread through the environment [27]. There are several ways in which ARB and ARGs in the environment can come in contact with humans: consumption of contaminated water and food, unhygienic conditions, and exposure to polluted soil and water. Animal-to-human and human-to-human transmission are also possible, depending on the sustained capability of ARB [28]. Overall in the last two decades, various efforts have been made to increase clinical surveillance of AR, which led to a better understanding of the occurrence of ARB and ARGs in clinical environments [4]. However, although there is increasing information on AR occurrence, fate and transport in aquatic environments, data is still limited, particularly in low and middle-income countries (LMICs). Due to the limited data availability, health risk characterisation is more difficult, and further translating that information into mitigative solutions has been slow [29, 30].

Fate and transport modelling can help understand the magnitude and differences in the source, transport of antibiotics with other pollutants into the environment, interactions with microbes, development and spread of resistance and associated health risks [31]. Here we review the factors and processes that could influence AR development, transmission, spread, fate, and transport in natural water environments. Our goal is to collate current knowledge to underpin new model development that will quantify transmission and spread of AR in aquatic systems. The review starts with an overview of AR in aquatic ecosystems and then discusses factors that potentially influence the fate, transport and in situ development of ARB, which are essential for developing efficient predictive models.

Table 1
Factors and processes that might influence Antibiotic Resistant Bacteria (ARB) and Antibiotic Resistance Genes (ARGs) fate and transport in the aquatic environments*.

Factor/Process	Short description
Antibiotic resistance	
Antibiotic residue	Antibiotic metabolites that are excreted after treatment in humans and animals
Antibiotic susceptibility	Sensitivity to antibiotics
Antibiotic cycling	Using different classes of antibiotics to reduce the bacteria that have already become resistant to the current drug classifications in use
Phage presence	Presence of phages that can hijack bacteria and influence the bacterial population
Bacterial interactions	Communication between bacteria in response to external stimuli (e.g., aquatic environments)
Bacterial evolution	Mutations or acquisition of new genetic material that allow bacteria to adapt and evolve in response to antibiotic selective pressure and environmental pressures
Bacterial growth	Growth of bacteria occurs in favourable environments, where the population size of the bacteria can be increased by cell division
Bacterial decay	Opposite of growth is decay, where bacteria perish in unfavourable conditions
Persistence	The existence of ARB or ARGs for longer periods in an aquatic environment
Degradation	The reduction of ARB or ARGs available in a particular aquatic environment
Propagation	The number of mechanisms or parameters influencing the proliferation of resistant bacteria
Survival rates	The percentage of the bacterial population alive after a certain period of time, represents the balance between growth and mortality rates, and the ability to survive in a particular water environment
Transduction	DNA transfer from one bacterium to another via bacteriophages
Transformation	Transforming by uptake of free genetic material
Conjugation	Transfer of resistant genetic material from one bacterium to another through direct contact

* Based on Andersson and Hughes [18], Davies and Davies [2], Holmes et al. [19], Lupo et al. [6]; Marti et al. [20], Martínez et al. [21], Munita and Arias [22], Sultan et al. [23], Taylor et al. [15] and Tenover [24].

2. Fate and transport of ARB and ARGs

2.1. Antibiotic resistance in aquatic environments

The complex array of AR sources, pathways, and sinks across the environments makes management solutions difficult [32]. There is increasing importance in research to understand the fate and transport of the ARB and ARGs in natural environments, particularly to develop better-targeted interventions and management [33]. Urban wastewaters, livestock, and aquaculture industries contribute significantly to AR spread. However, the relative impact of each sector is unclear due to a general lack of adequate data [34]. A word cloud (bibliometric) analysis illustrates that the majority of the modelling work on antibiotic resistance spread is concentrated in clinical environments in contrast to the natural aquatic systems where research is only emerging (Fig. 1, using VOSviewer; Table S1) [35]. However, the research on antibiotic resistance modelling has grown tremendously over the last decade, especially in natural environmental settings, because of the topic's increasing interest and health emergency (for additional information, please see Supplementary section S1).

For example, hospitals, cities and farms (livestock, aquaculture and crops) discharge increasing loads of ARB and ARGs to water bodies [36–40]; often, resistance develops at the source [41,42] (for additional information, please see Supplementary section S2). Once in receiving waters, ARB and ARGs can be further transmitted and spread [43], and ARB populations can grow, change, and/or decay. Indigenous bacteria already present in the water environment can also develop resistance when interacting with antibiotics, ARB and ARGs released into this environment [44,45]. ARGs can persist and reside in the environment for longer periods, although their importance to health risk is unclear because genes only imply the “potential” for AR, not actually expressed AR [30]. Regardless, evolutionary drivers appear to have made them ‘easy to get and hard to lose’ pollutants, which makes them particularly worrisome [20].

Non-resistant bacteria in aquatic environments can also gain resistance because of selection pressure caused by numerous trace organic compounds, including antimicrobials, heavy metals, and non-antibiotic pollutants released into the environment [41]. Antibiotics become particularly important where local concentrations are very high, and whilst this does not imply trace amounts of antibiotics have no effect, the relative effect might be small compared to other factors. A wider cocktail of antibiotics released directly into the environment along with ARBs and ARGs [46]. To date, 57 antibiotics have been identified in lakes around the globe, which can conditionally select for AR within lacustrine environments [47]. In addition, possible incubators for bacteria in water environments, the mixture of antibiotic concentrations, disinfectants, other pollutants and broader interactions in the environment can influence the evolution and selection of AR and multidrug-resistant (MDR) bacteria, which is a further threat to human health [19].

Further, ARB and ARGs are becoming ubiquitous in aquatic environments, including wastewater treatment plants [38,40,48], rivers and streams [49–51], lakes and reservoirs [52,53,47], estuaries and coastal environments [54,55], soil and groundwater [56–58]. Around the world, riverine aquatic systems are the principal water bodies that receive treated and untreated wastewater [59], which act as transport carriers for ARB and ARGs [60]. While there is limited knowledge about AR evolution in aquatic environments, however, the mechanisms that control AR in controlled environments are well understood. We can use this information to acquire a more nuanced understanding of ecological, environmental and hydrological processes that influence the evolution, spread, persistence and survival of ARB and ARGs in natural aquatic environments [15,61].

2.2. Fate and transport of ARB and ARGs in different systems

Developing a working AR water quality model has the potential to provide information and enable us to predict the impacts due to changing conditions and help us in decision-making. However, the model must be reliable by recapitulating the empirical-observed patterns of occurrence of resistance. This section reviews some of the important contributing factors that affect the fate and transport of AR in different environments. Not all potential factors are included here for the sake of space, i.e., we only focus on contributors present in almost all landscapes. Other factors are summarised in detail in recent UN guidance on environmental AMR [26]. The environments are categorized for discussion based on the similar kinds of fate and transport mechanisms of AR.

2.2.1. Wastewater treatments plants (WWTPs)

The natural occurrence of AR in the environment is comparatively low, and several studies have shown high levels of ARB and ARGs linked to point and non-point anthropogenic sources, especially polluted waters associated with wastewater [62,25,3]. The main pathway for ARB and ARGs to enter aquatic environments is via domestic wastewater, including wastes from hospitals and other community sources [43,46,48]. Thus, inadequately treated wastewater and WWTPs can be important conduits of ARB and ARGs to local environments [63]. Untreated domestic wastewater is potentially the most significant source of AR to the environment on a global scale [26], and almost any form of wastewater treatment is betting that nothing [64]. However, for the purpose of this review, focus will be placed on WWTPs and their contribution to ARB and ARG releases to aquatic environments.

Secondary biological WWTPs can contain a variety of species of the dense bacterial population and abundant nutrients, which appears to provide a suitable environment for the co-mingling of ARB and ARGs [65,66,48] (for additional information, please see Supplementary section S3). However, it is not certain the extent to which new AR develops in WWTPs or whether WWTPs are simply a conduit of AR that has developed within sewage networks, such as from hospital sources [67].

There are successful models for the removal of biomass, phosphorous, carbon oxidation, nitrification etc., in WWTPs [68,67,69,70], but the models for population dynamics of microorganisms in the presence of antibiotics and ARGs considering evolution and spread of resistance in WWTPs are in the early stages of development. A few studies have attempted to simulate antibiotic degradation dynamics in activated sludge reactors to understand the degradation kinetics of antibiotics, but only a few included the AR dynamics, and few consider recent knowledge of AR and HGT prevalence in WWTP sludge and liquid effluents [71–73].

A model that formulates a mechanistic or holistic understanding of the evolution and spread of AR in WWTP would greatly help to understand resistance in microbes and their geochemical interactions in natural aquatic systems. Recent work has shown that AR released from WWTPs may be most impacted by clarifier performance [67], which may be helpful for new models that aim to calculate the AR loads before releasing or applying effluent and sludge into the natural environments.

2.2.2. Soils and agricultural land

Soil is the largest reservoir of diverse microbial communities and intrinsic AR [74,75]. However, applying animal manure as fertilizer, use of wastewater for irrigation, use of sludge from WWTP as a soil conditioner, and direct deposition of faecal matter by livestock and wild animals can increase both the diversity and prevalence of AR in soils [76–79], and mirrors wider increases in AR across the One Health spectrum [80] (for additional information, please see Supplementary section S4). Once the ARB and ARGs enter the soil environment following rainfall/irrigation events, nutrients, antibiotics and bacteria are mobilized and transported through surface runoff and infiltrating water into surface water bodies or groundwater. Hence, ARB and ARGs



supplementary material section.

are released from their original source, partitioned into particle-associated or soil solution fractions, and routed through or across the soil via different hydrological pathways. Effective modelling (both process understanding and prediction) requires a good understanding of the survival of ARB and ARGs in manure and soil, sorption in the soil reservoir, transport in the soil column, release of bacteria from biosolids, soil moisture and texture, cell growth and morphology, etc. [81]. The processes of ARG fate in the soil are complex and dependent on the relative growth-decay of bacteria carrying the ARG, microbial predation and other food web effects on ARB fate, and the relative tendencies for HGT [82,83].

The vertical transport of bacteria in soil columns has been studied by injecting bacteria suspensions or wastewater into soil columns or releasing bacteria from manure to the soil surface [84-86]. Several soil column experiments were conducted to gain more information and evaluate manure application's influence on the subsurface fate and transport of ARGs [86-88]. Barrios et al. [89] summarized the optimized parameters for the attachment-detachment model for the vertical fate and transport of ARGs in soil and reported a significant change in the first-order attachment coefficient due to manure application and no significant change in first-order detachment coefficients. Other studies as well corroborated and reported that there is a minimum long-term impact of field application of manure on the leachate water, suggesting that ARGs get attached to soils instead of leaching to the groundwater, and the soil is the main source of ARG in groundwater [86-88]. Another interesting observation related to ARGs released from manure is that they occur in extracellular DNA, but in livestock manure, they majorly exist in intracellular, i.e., their bacterial hosts. Therefore, the concentration of ARGs depends on the fate of the bacterial host carrying them, which depends again on the release kinetics of manure-borne microorganisms by their cell characteristics, their distribution in manure particles, and microbial ecological factors [90-93].

2.2.3. Sediments and groundwater

ARB and ARGs are released into the environment via myriad sources that can enter groundwater systems [94,95]. The role of antibiotic

concentrations in groundwater over the dissemination of ARB and ARG is not yet researched extensively; however, it has been suggested that groundwater might be considered an important reservoir for ARB [56]. Various studies have shown that mechanisms, such as cross-resistance due to other selective pressure, influence of bacterial community and co-resistance might play an important role in ARG evolution and persistence. But limited research has been done on the time-dependent dynamics of ARB and ARGs in the aquatic environment, which can reflect more on the fate and transport of ARB and ARG in sediments and groundwater.

Faecal indicator organisms (FIOs) such as *Escherichia coli* are routinely observed in streambed sediments, banks, and soils [96]. There are models for FIOs released with resuspended sediments and their retention in transient storage. Whereas little is known about the influx of FIOs from sediment in groundwater, a high concentration of FIOs was observed during the baseflow in some waterbodies. A substantial percentage of the *E. coli* population can originate from the streambed, which indicates that sediments may act as a reservoir and might be brought to the stream through groundwater [97,98]. Hence, this information could be referred to and applied for modelling scenarios, as FIOs are usually carriers of ARB and ARGs [99,100]. Genetic source tracking has shown that this is the case in wastewater-impacted river sediments where significant abundances of original faecal ARB are the sediment-host of ARGs and MGEs, such as class 1 integrons [67,101]. The material exchange of surface water and shallow groundwater through sediment layers (hyporheic exchange) might result in the transmission of ARGs to the groundwater. Therefore, it is important to explore the fate and transport of ARB and ARGs considering surface water-groundwater interactions [102].

2.3. Environmental factors and processes that affect fate and transport

To model an environmental system, it is necessary to understand the spectrum of climatic, hydrological, and ecological processes and factors (precipitation, water flow dynamics, kinetics, temperature, residence times, etc.) that can influence the movement of bacteria from point A to

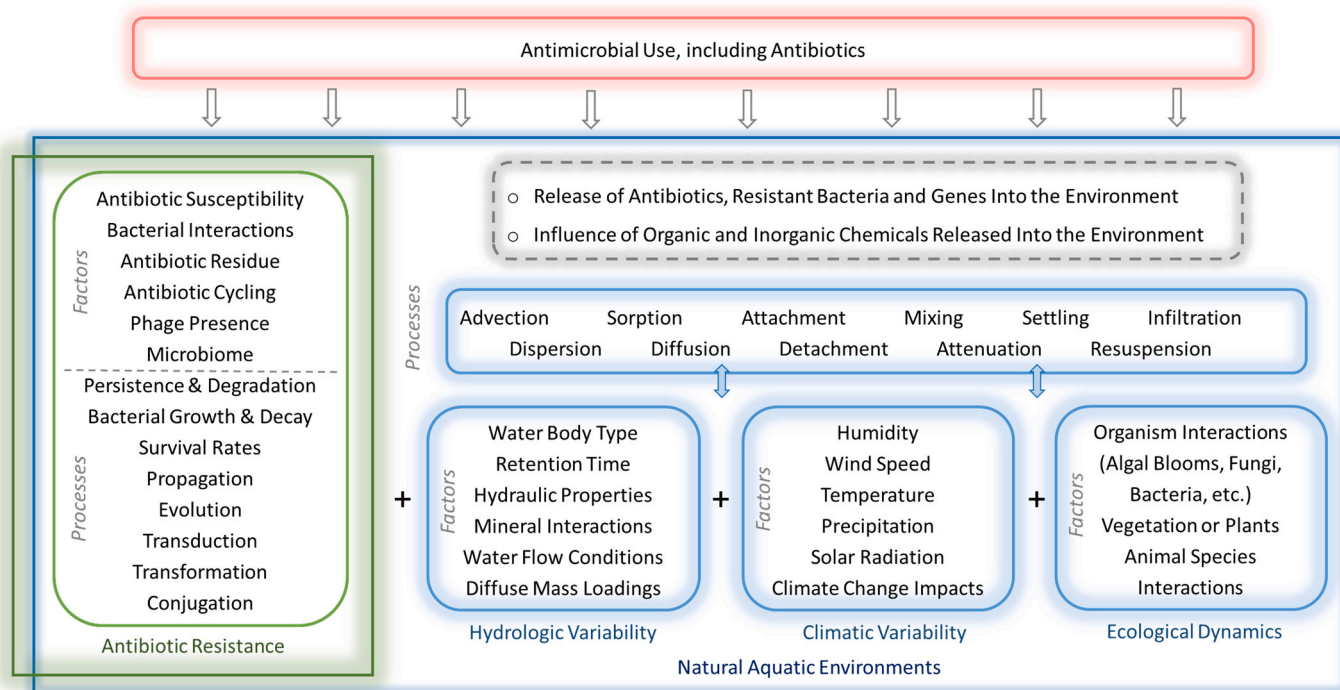


Fig. 2. Comprehensive view of various factors and processes that might influence Antibiotic Resistant Bacteria (ARB) and Antibiotic Resistance Genes (ARGs) fate and transport in the aquatic environments.

point B in a water body and their capacity to survive, evolve or die in aquatic environments. A comprehensive list of these factors and processes is presented in Fig. 2 and further explained in Table 1. ARB can grow in many diverse environments because they often tolerate greater chemical stress, and ARGs can remain in potentially extreme environments from several days to several months [103,3,60,15,45]. Furthermore, MDR bacteria are often more prevalent in the water when compared to other environments because defence against multiple stressors has an ecological advantage [104].

A relatively significant portion of ARB in the river seems to be correlated with different anthropogenic activities in its watershed [105–108]. Therefore, an increasing number of studies targeted the presence and the persistence of ARGs in water and soils that are exposed to human activities [52,109–112], and often due considering spatial and temporal variability, which are assessing exposure risk places with significant seasonal climatic variations [113,114]. While these processes may need to be considered by water quality microbial fate and transport models, most studies in the literature do not consider or intentionally avoid any hydrological or meteorological disturbances.

A rainfall event, especially after a long dry season, might lead to an increase of ARG in surface waters due to accumulation and following runoff with combined delivery of nutrients and microbiological aggregates. Due to rainfall events, the biofilm over the soil gets dislodged and fractionated when it reaches the surface water body, which is a potential source for resistance determinants. There has been a strong correlation between infectious diseases and hydrological events or disasters [115, 116]. However, the research so far has focused on the impact of hydrological variability concerning bacteria, with only a few exceptions. Cesare et al. [117] explained that moderate rainfall events could affect ARG dynamics in the resistome of a large subalpine river. The increments in the ARG abundances were noted during the rainfall, along with an increase in concentrations of total phosphorous, N-NH_4 , and microbial aggregates [118]. These studies reported a higher proportion of potential ARB, which indicated a general shift in the microbial community composition due to moderate rainfall events.

Pathogen survival and growth depend on the type of aquatic ecosystem. A pathogen that persists and grows in a riverine environment might not survive in groundwater due to differences in temperature and mineral conditions [95]. The long retention times and facile water exchange conditions can increase HGT and AR selection and can be critical hydraulic conditions that may drive AR in lacustrine ecosystems [47]. In contrast, short retention times in high-flow periods can increase the ARG abundance in ecotones [119]. The determinants of AR can also be influenced by river/stream dilution and pollutant decay over long distances [120]. Clear evidence suggests bacterial movement can be slowed down in flowing waters because of fluid shear [121], which may affect bacterial fitness in a highly hydrodynamic environment. However, the suspended sediments in rivers and streams can be reservoirs for ARB, and through hydrological flows, these sediments can be deposited in reservoirs and lakes. ARB can be more prevalent in stagnant water bodies, which can help a weaker pathogen to evolve, persist, and disperse to the broader ecosystem [27].

Mineral composition, sediment or soil grain size, and grain size distribution substantially influence bacteria deposition in porous media [85]. In addition, the bacteria cell characteristics might play an important role in fine sand environments, and transport processes have a larger impact on bacteria in coarse sediments. Understanding the impact of various hydrological characteristics (advection, dispersion, diffuse mass loadings, retention time, etc.) on different ARB and their ARGs is still in the early stages of research. Further research exploration will enable a deeper understanding of the survival capability and fate of certain types of ARB in different aquatic environments.

The occurrence of ARB in aquatic systems is strongly influenced by seasonal climate variability. For instance, higher concentrations of antibiotic-resistant *E. coli* were observed in wastewater effluents after the rainfall events in Georgia, USA [122]. Many studies have identified

rainfall as one of the key factors influencing the abundance of ARB and ARGs in aquatic environments [118,54,117], especially in monsoonal climates [113]. This may result from the leaching of the pathogens from places where they are accumulating and multiplying. Levels of pathogenic bacteria in wet seasons can be tenfold compared to those in dry periods of the year, and some of the ARGs are more abundant in wet conditions [118,123]. With the change in wet and dry conditions, there might also be an increase in the pathogen bacterial load and transport conditions from soil media to surrounding water bodies [124]. Precipitation might flush ARB and ARGs from surfaces and soils into water bodies, and higher flow velocities under high-flow conditions can increase sediment resuspension and ARG transport, including ARGs stored and accumulated during the dry season [113]. Unless those surfaces were already recently flushed, in which case increasing precipitation might not have much impact on growth and may actually dilute the concentrations. Jeon et al. [125] evaluated the fate and transport of bacteria under changing climate conditions and pointed out a substantial increase in the proliferation of bacteria due to an increase in temperature and precipitation. Higher temperatures and more moisture may result in favourable bacteria growth.

Climate change affects disease patterns and antibiotic use on a global scale, which will influence ARB spread and evolution in aquatic environments [126]. The complex changes to biogeochemical cycles associated with climate change and their impact on AR can be myriad and vary from region to region [127,128]. Climate change may not influence some processes, but it certainly changes the locations where those processes can occur. It may trigger the migration of AR pollution to other regions that previously did not have the best or most suitable climatic conditions for ARB evolution. Antibiotic use might become more necessary in regions that get warmer and moist. Pärnänen et al. [40] identified that in countries where animal agriculture is practised, those with lower atmospheric temperatures have a lower abundance of ARGs in wastewaters. An increase in temperature range with climate change (stretching the limits of minimum and maximum temperatures) could force local microbial populations to adapt and contribute to the rise in AR [129]. With climate change, it will be of growing importance to consider the climatic effects of temperature, humidity, rainfall, etc., on the fate of ARB and ARGs.

The ecological dimension in assessing ARB and ARG fate and transport is multi-layered and complex with species and environmental interactions [130]. When we speak of ecology, both microbial ecology and environmental ecology have diverse impacts in disseminating resistance to the natural environment. Microbial ecology explains the interconnectedness of different resistant or non-resistant bacterial species with one another and their relationship with the surrounding environment. Environmental ecology is where resistant/non-resistant bacteria interact with different organisms in the broader environment. This innate behaviour of ARB and the ecological functions of watersheds can become crucial in determining fate and transport. It would be beneficial to look into emerging ecological theories (e.g., ecological coexistence theory) to understand the hydroecological dimensions of AR and corresponding microbial competition, which might be helpful in elucidating the ARB proliferation and decay rates [131]. Bacteria with a capacity to survive in harsh environments or low-nutrient systems can develop resistance at a higher rate when compared to bacteria that grow in nutrient-rich or controlled environments [132]. However, diverse ARB are transmitted into even more diverse natural ecosystems, and their survival may be primarily driven by whether or not they find a local niche in the native community in the receiving water [133]. Plants and vegetation also might play a crucial role in supporting the microbial and environmental ecological links, especially in wetlands and lacustrine ecosystems. A recent study of the *Sphagnum* moss microbiome revealed that a novel set of ARGs and multidrug resistance observed in bog ecosystems [134]. Bacteria can also ‘hitchhike’ to get transported at the impenetrable density gradients using aquatic organisms such as zooplankton [15,135]. Ecological interactions between bacteria in

freshwater and zooplankton play a critical role in the persistence of ARGs in aquatic environments. Planktonic crustaceans like *Daphnia* can be potential carriers of ARB in freshwater environments [136]. In aquatic microbiology, viromes can also act as genetic reservoirs to spearhead the AR in water systems [137]. Climate change also influences the functional ecology and aetiology of ARB within aquatic ecosystems [138,139]. We need a more precise understanding of interactions between organisms, vegetation, plant and animal species when it comes to understanding the ecological dynamics of different resistant microbial communities in aquatic environments.

2.4. Dynamics of susceptible and resistant bacteria in aquatic environments

The presence of antibiotics in the environment sometimes plays a role in the persistence and mobilization of resistance [27,140,141], although the relative importance of such chemical exposures vs other factors in the transmission and spread of AR is under debate. Novel ARGs can emerge from the vast population of bacterial cells through mutations, rearrangements, and HGT, but experimental evidence of how it happens in situ is lacking. Speculation has been that AR enrichment will not happen without selection pressure, as resistance factors are generally associated with some fitness costs [27]. However, factors other than carrying an ARG may dominate the survival of such strains in the absence of very strong selective pressures, such as higher concentrations of antibiotics or heavy metals. Much work is still needed to clarify the dynamics of AR selection, which may, in fact, be less impacted by in situ antibiotics themselves but controlled by other factors, such as specific allochthonous ARBs (e.g., from wastewater) might “fit” and survive in the local autochthonous microbial community in which it “invades” [133].

Regardless of the original reason for selection, over time, compensatory mechanisms appear to maintain ARGs even in the absence of selection pressure. Resistant and susceptible bacteria are always in head-to-head competition in an aquatic environment for limited resources in the absence or presence of selection pressure [142,143]. If a bacterium carries an ARG, a fitness cost may alter its growth rate. However, how that impacts the survival of the ARG host will depend on the strain’s potential niche in the given environment. In some cases, susceptible bacteria will outcompete the ARB when selection pressure is absent, but in other cases, an altered growth rate may play no role in the selection, such as if the ARB was being sustained in the community in a K-strategy role (i.e., scavenger). This leads to many scenarios requiring more investigation into what happens in the field vs lab settings, where most data exist on AR dynamics. Existing lab results have not considered wider community effects, which may dominate AR selection in natural aquatic ecosystems.

2.5. Mass balance equation

Water quality parameters are usually measured in concentration units or mass quantities. Mathematical models of the fate and transport of a contaminant usually account for the processes and mechanisms to calculate its concentration. Thus, it is easier to determine the fate of any parameter in natural water using a mathematical model. The key elements for the mass balance of ARB and ARGs: gene copy concentrations require knowledge of gene loadings, initial baseline ARB and ARG concentrations, the growth rate of ARB, HGT rates, decay rates of ARB and ARGs and transport mechanisms in different compartments of the environment. Hence, a mass balance is simply accounting for mass inputs, outputs, reactions, and accumulation as described below:

ARB/ARG accumulation within a control volume

$$= \text{ARB/ARG inputs} - \text{ARB/ARG outflows} + \text{HGT} \\ \pm \text{change in ARB/ARG decay.} \quad (1)$$

The pathway of ARB and ARGs into the water environments is usually through point or non-point sources such as WWTP, runoff from agricultural fields, contaminant leaching and recharge, and direct interaction of humans and animals with the water system. Therefore, it is crucial to gather detailed information on the fate and transport of ARB and ARG in each compartment of the watershed to understand their corresponding fate and transport in natural waters.

3. Modelling ARB and ARGs in the aquatic environments

3.1. Modelling fate and transport of bacteria: current state of the art

Microbial water quality modelling started with drinking water and then recreational water and irrigation water and was further developed for whole watersheds [144]. These models are usually physically-based and mass balance models based on hydrologic transport, e.g., Streeter-Phelps equation [96]. Research on understanding the fate and transport of microbial communities has advanced in the last four decades in both controlled environments, including laboratory systems, wastewater treatment plants, etc. [145,146] and natural environments, including rivers, lakes, aquifers, etc. [147,148]. Practical research methods have been developed to study and model the fate and transport of environmental bacteria in aquatic environments [149,150]. Modelling bacteria transport is essential for providing a solid foundation for developing a transport model for ARB in the environment.

Many studies have been conducted in lab settings to understand the biological processes affecting waterborne bacteria population change rates when interacting with river or lake sediments or during infiltration. Bacterial population growth models have evolved since the 1960s to predict survival, filtration, attachment, and die-off rates. Several studies have employed continuous injection of synthetic water to study straining processes [151,152], filtration [153,154], and kinetic attachment-detachment processes [155,156]. These models are grouped into mechanistic models describing the processes bacteria undergo when interacting with water sediments or soils (Table 2). The process rates and parameters vary based on the flow rate of the water passing through the sediment/soil, type of sediment/soil, grain size of the sediment/soil, and bacteria’s size. In addition, other studies have focused on modelling bacterial transport, where treated wastewater with spiked bacteria concentrations is injected into soil/sediment columns, and the resulting breakthrough curves are analysed [157]. Integrating these factors in pathogen fate and transport modelling is often complex, and the detailed understanding of pathogens in various water environments is still a big challenge [96]. A major challenge in modelling AR is separating or identifying various simultaneous processes that could occur in the vadose zone and water sediments and considering how complex ecological interactions, including food web effects, influence predictions. While microbial population models are still in the relative development phase, over the years, several models have been developed to deal with different microorganisms in various water environments [149,124,96].

Microbial fate and transport modelling in different environmental systems depends heavily on bacterial decay, die-off rates and behavioural characteristics, hydrological pathways, land use and climate change [124,150]. Process studies on a lab scale have helped the development of large-scale models, which has led to the development of software for reactive transport modelling of bacteria on a larger scale. The Soil and Water Assessment Tool (SWAT), which includes microbial fate and transport modelling, has been widely applied to various water systems and watersheds across the world [161–163]. A new bacteria subroutine was also introduced recently for the SWAT for modelling surface water systems [164].

Similarly, the United States Department of Agriculture developed another recent model, Microorganism and Solute Transport in Streams (MaSTiS), for modelling concentrations of *E. coli* fate [165]. Also, the fate and transport of viruses in unsaturated soil have been modelled in

Table 2

Summary of previous investigations focusing on bacterial and colloid transport in saturated porous media using the advective dispersive equation.

Experiment objective	Process Studied	Corresponding equation	Parameters	Reference
Effects of colloid shape on straining kinetics and quantify the effects of colloid and collector size on straining rates	Straining	$\frac{\rho_b}{n} \frac{\partial S}{\partial t} = k_o C_e \frac{S}{\lambda}$	k_o : clean bed straining rate λ : straining co-efficient	[152, 158]
Influence of straining on colloid transport and fate	Straining	$-\frac{\partial(\rho_b S)}{\partial t} = \theta_w k_{att} C + \rho_b k_{det} S$ $k_{att} = \frac{3(1 - \theta_w)}{2d_{50}} \eta \alpha v$	k_{att} : first-order colloid attachment co-efficient k_{det} : first-order colloid detachment co-efficient d_{50} : median porous medium grain diameter η : collector (porous medium) efficiency α : colloid sticking efficiency v : pore water velocity	[159]
Impact of bacterial interactions on transport and sorption processes. Understanding of bacterial migration as it pertains to bacteria-bacteria and bacteria-sediment interactions	Attachment detachment	$\frac{\rho(1 - \varepsilon)}{\varepsilon} \frac{\partial(\rho_b S)}{\partial t} = -k_{irr} \left[C + \frac{\rho(1 - \varepsilon)}{\varepsilon} S \right]$	$\rho(1 - \varepsilon)$: dry bulk density of the porous media k_{irr} : irreversible attachment co-efficient	[160]
Fate and transport of <i>E. coli</i> bacteria in the vadose zone	Irreversible attachment, decay	$R \frac{\partial C}{\partial t} = D \frac{\partial^2 C}{\partial x^2} - v \frac{\partial C}{\partial x} - \mu C$	μ : overall first-order decay rate R : dimensionless retardation factor from irreversible adsorption	[157]

the VIRTUS software [166]. Because of their sophisticated modelling structure, using similar models might enable predictive modelling of ARGs in the natural environment. While these large-scale models give a more realistic representation of the environmental systems, they require a lot of data and parameterisation for the results to be reliable and to be used for risk assessment or policy development [167,168]. A significant constraint on model development is the generation of data. Data from the monitoring campaigns in natural aquatic environments like canals, wastewater treatment plants, and rivers would enable quicker and more accurate models to be developed to understand the spread of pathogenic bacteria, including ARB and ARGs, in aquatic environments [53,169,58,170,171].

3.2. Current modelling efforts of ARB

Various modelling efforts to understand AR development and spread have focused on understanding transmission in hospitals, communities and at animal-human interfaces [172]. Mathematical models like population level, risk management and dose-response have been developed to understand the AR infection rates or risks [173-175]. Beyond this, AR modelling has developed slowly compared to other infectious disease modelling, especially in the host and within the environment [176,177,172]. Gothwal and Thatikonda [178] developed a site-specific mathematical model for understanding the transport of fluoroquinolone-resistant bacteria in a river. More case studies like this are required to help develop general models suitable for ARB transport in various aquatic environments. Various other analytical and numerical models have been developed to precisely understand the fate of certain ARB [179-182] (Table 3). Combining microbial source tracking methods for ARB and ARGs or surrogate markers, such as dissolved oxygen (DO) or ammonia, for more generic modelling in environmental systems has shown promise [114,183].

Further research into the extrapolation of such models to fit into the natural environment could enable successful modelling of the environmental resistome, including ARGs, transported into and/or from the environment [31] (Fig. 3). In addition, statistical modelling, including principal component analysis (PCA) and microbial community analysis, can help clarify the correlations between anthropogenic source and in situ resistance patterns. Predicting the persistent and non-persistent pathogen's fate and transport in the water environments itself is difficult and complex. Considering Figs. 1, 2, 3, and 4, the number of sources combined with the complexity of fate and transport increases to a greater extent when a bacterial population becomes resistant to one or

Table 3

Research studies from the literature that performed water quality modelling and evaluated antibiotic resistance in natural aquatic environments.

Research article	Study area	Model description
[181]	Poudre River in the USA	Based on combining the existing mathematical models to evaluate tetracycline-resistant bacteria occurrence
[180]	Poudre River in the USA	Mathematical model to determine the heavy metal co-selection on tetracycline-resistant bacteria occurrence
[178]	Musi River basin in India	Mathematical model to evaluate fate and transport of fluoroquinolone-resistant bacteria
[184]	Outdoor experiment of a hydraulic channel (U-shaped with 1% slope; Sele River, Italy)	Modelling the fate of antibiotic-resistant <i>E. coli</i> using the numerical simulation of fluid flow of an artificial stream
[179]	Musi River basin in India	Mathematical model employed stochastic partial differential equations for assessing the transport of fluoroquinolone-resistant bacteria
[182]	General model for wastewater receiving waters	Finite difference model to predict the fate and transport of ARGs in the (river) downstream of a wastewater treatment plant
[185]	Musi River basin in India	Mathematical model to assess the fluoroquinolone-resistant bacteria considering nutrients and heavy metals as variables
[31]	Global and conceptual model	Holistic conceptual modelling framework to evaluate the ARB and ARGs in watersheds
[186]	Dutch-German transboundary Vecht River	Modelled resistant <i>E. coli</i> using a River catchment model combined with mass balance approach (GREAT-ER)

several antibiotics – understanding the underlying mechanisms leading to the evolution of resistance and transmission is complex itself. It becomes a greater challenge to model the processes that lead to the evolution and transmission of resistance in natural environments (Fig. 3). When a bacterium becomes resistant, its decay and die-off rates may change. If the carriage of ARGs creates a fitness cost, then ARB may be

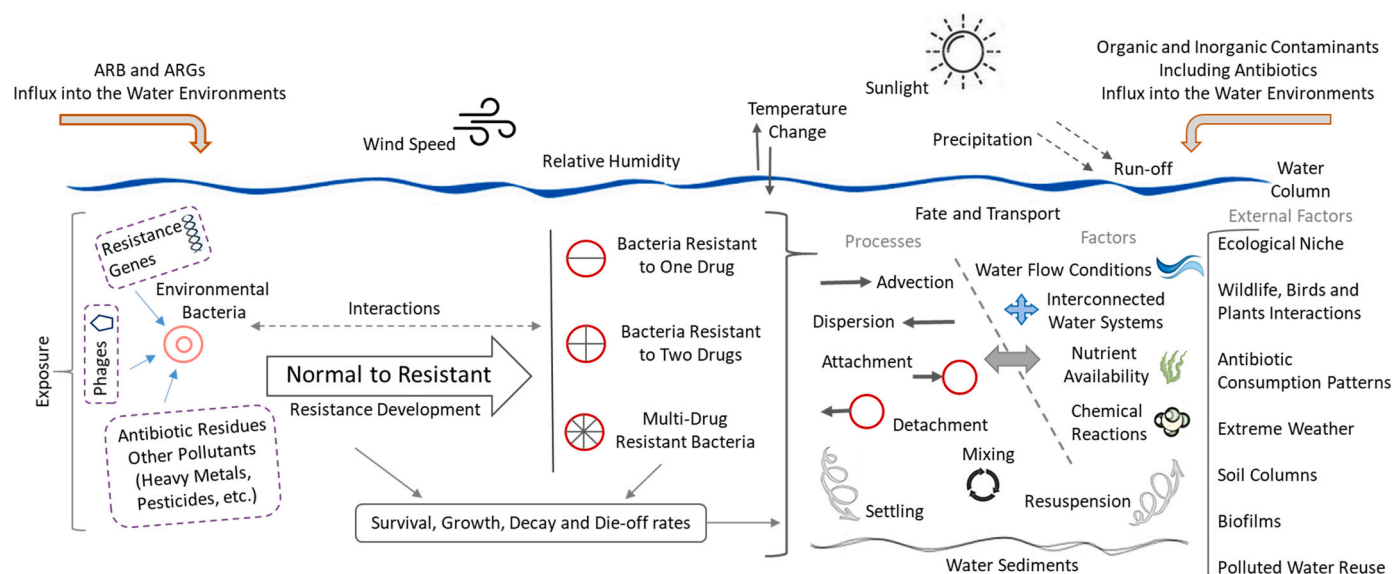


Fig. 3. Modelling complexity to tackle in a typical watershed to address or evaluate the antibiotic resistant bacteria (ARB) fate and transport processes.

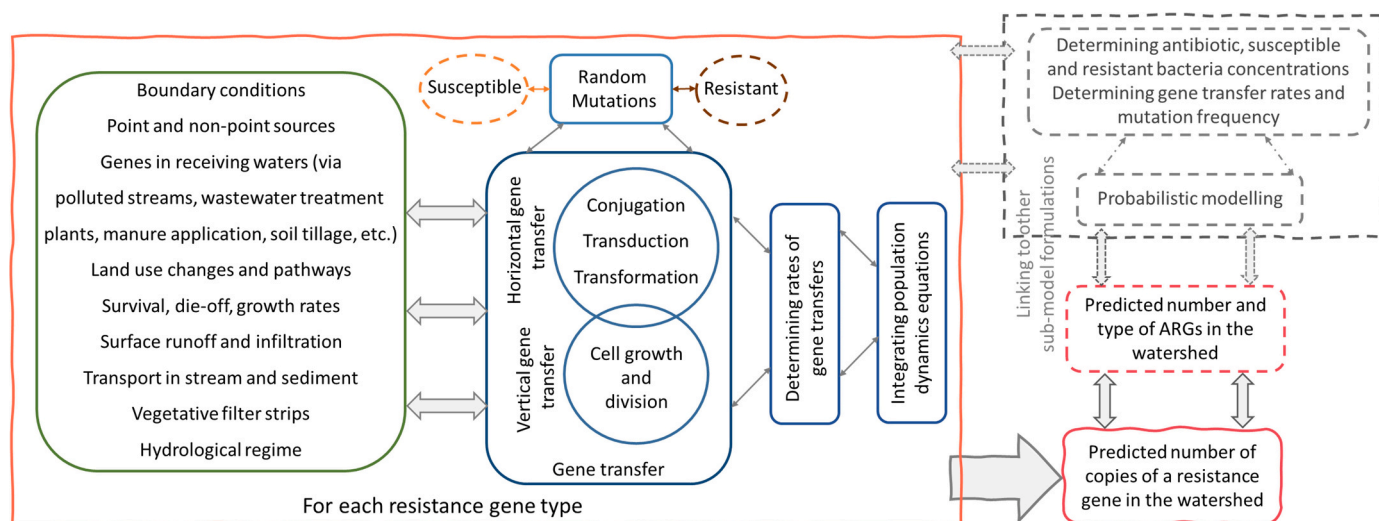


Fig. 4. Conceptual flow diagram of the antibiotic resistance genes (ARGs) fate and transport model module for evaluating the number of resistance gene copies in a watershed.

selected out of the waterborne population. The fate and transport of an ARB may therefore be affected by:

- Environmental survival of ARB in different bodies, with respect to hydrological, climatic, water chemistry (e.g., DO or oxidation-reduction conditions) and ecological interactions.
- Further resistance development in the aquatic environment itself in response to antibiotics, heavy metals, and other pollutants in water.
- Resistance development due to the possible acquisition of freely available ARGs in the environment presumes bacteria take up free DNA in a mobilizable manner.

3.3. Gene transfer mechanisms

Modelling of bacteria can be extended to include various gene transfer processes that are responsible for the development and spread of resistance in the environment (Fig. 3). Resistance development has been modelled using simple probabilistic models, such as those that quantify the probability of a mutation to occur given the 'appropriate conditions',

such as by selection due to antibiotics, heavy metal ions or other factors [187]. Different environmental constituents can influence ARG abundance [103]. Antibiotics might have the least impact on ARG dynamics in the natural environment, but they appear to indirectly influence ARG profiles by altering HGT frequencies and bacterial community changes [62,102].

Other factors also affect HGT rates, such as oxidation-reduction conditions, microbial predation, and the prevalence of biofilms [93]. Regardless, HGT is a principal process that defines the ARB evolution, and reliable quantitative data on HGT rates and frequencies is a significant research gap for modelling, even in clinical settings [177]. Some models have been developed for HGT occurring in human or animal guts [188,189], which might be useful for adopting the equations for HGT mechanisms in natural environments (Table 4). Previous modelling of HGT has focused on simple differential/analytical models [61,190–192]. It was uncertain how different gene transfer processes, such as conjugation, could be modelled to predict the ARGs and their corresponding ARB growth in aquatic environments. It would be interesting to look into ARG transfer rates during conjugation in clinical environments and

Table 4

Equations representing the possible description of conjugation: one identifiable horizontal gene transfer process from the mass action model developed by Levin et al. [194].

Variable description	Processes and variables	Growth	Conjugation
Recipient bacteria	$\frac{dR}{dt}$	$\Psi_{\max} \cdot \frac{S}{Q+S} \cdot R$	$-\gamma_{\max} \cdot \frac{S}{Q+S} \cdot R \cdot (D+T)$
Donor bacteria	$\frac{dD}{dt}$	$\Psi_{\max} \cdot \frac{S}{Q+S} \cdot D$	
Transconjugant bacteria	$\frac{dT}{dt}$	$\Psi_{\max} \cdot \frac{S}{Q+S} \cdot T$	$+\gamma_{\max} \cdot \frac{S}{Q+S} \cdot R \cdot (D+T)$
Nutrients	$\frac{dS}{dt}$	$-\Psi_{\max} \cdot \frac{S}{Q+S} \cdot (T+D+R)$	$-\gamma_{\max} \cdot \frac{S}{Q+S} \cdot R \cdot (D+T)$

Parameter description: R, D, T, S: Concentration in $\frac{\text{CFU}}{\text{mL}}$ of the recipient, donor, transconjugant bacteria, and nutrients ($\frac{\text{mg}}{\text{L}}$); $\Psi_{\max}$: Maximum growth rate of the bacteria (min^{-1}); Q: Half-life of the growth and conjugation process ($\frac{\text{mg}}{\text{L}}$); $\gamma_{\max}$: maximum conjugation rate (min^{-1}).

determine how it might be translated into natural environmental settings [193]. One novel approach is developing plasmid-labelled marker-strains and introducing them into different aquatic environment simulations, which Jong et al. [93] did with plasmid- and host-labelled *E. coli* to quantify HGT frequencies in different wastewater treatment environments.

Incorporating resistome's behaviour in different aquatic environments can make it very difficult to model gene transfer mechanisms. One empirical approach is to study the rate of plasmid transfer in bacteria, which has been done using the so-called end-point method, where the bacteria conjugation rates are modelled using simple ordinary differential equations [195]. While these models do not accurately represent the conjugation rates for most bacteria, usage and improvement on these models could be a starting point to model plasmid transfer in both lab and natural environments. In addition, the Hill Equation has been used to understand the pharmacokinetics of the relationship between the predation of antibiotics and bacteria [196]. Development of these models using data from a lab scale and eventually with field monitoring data would enable us to predict the emergence and transmission of AR.

Other models that quantify factors that impact the transfer of AR to humans could be vital in developing the risk assessment linked to the fate and transport model [197]. The main constraint is that it would be more complex to understand the gene transfer rates and mechanisms in natural aquatic environments. Therefore, we need to look for more robust methods to model the complexity of gene transfer mechanisms. The mass action model that Levin et al. [194] described might be useful for modelling antibiotic emergence and processes in aquatic systems (Table 4). Assumptions that were made for the model: The predominant method of gene transfer is through conjugation; it is assumed that the process of conjugation is spontaneous and takes place only with the donor and the recipient bacteria; once the process has taken place, the transconjugant can immediately express resistance; there is no decay or die-off of the bacteria in this system; the growth rate of the donor and transconjugant bacteria are the same and is represented using the Monod growth curve. Although most of these assumptions will not hold true in natural systems, the model provides a starting point for teasing out what can and cannot be modelled from fundamental presumptions.

3.4. ARG fate and transport model module schematics

The supposed model module could comprise a set of ordinary differential equations representing the resistant and susceptible bacteria population changes. The model should combine the concepts from the existing models for the fate and transport of antibiotics, bacteria, and

resistance genes transfer [96,179,178,140,180,181,190,172,189]. The hydrological processes such as advection, dispersion, settling, and resuspension of genes should be adopted from the established methods and existing microbial water quality fate and transport models. However, the potential for AR evolution, HGT, in situ microbial cell densities, and the influence of water quality conditions, such as DO, are additional elements needed in modelling. It is suspected that sub-inhibitory antibiotic concentrations can selectively promote AR's evolution in many environments [198,199]. However, the relative importance of low antibiotic concentrations on in situ AR selection has only been shown in a few field cases [200], and other factors like HGT may be more important in natural environments [201].

Overall, the model module consists of the variables such as susceptible bacteria (S), resistant bacteria (R), ARGs, total population of bacteria in the aquatic environment (N), antibiotic concentrations (A), heavy metal concentrations (M), the concentrations of nutrients, dissolved oxygen (DO) and dissolved organic matter (DOM); HGT rate via conjugation (β_c); HGT rate via transformation (β_f); HGT rate via transduction (β_d); The growth rate of susceptible bacteria (μ_S); The death rate of susceptible bacteria (D_S); The growth rate of resistant bacteria (μ_R); The death rate of resistant bacteria (D_R); Mutation (α); Segregation or un-mutation (γ); Rate of disintegration of bacteria (lysis rate); Metal inhibition rate of constant (k_i); The half-life constant (K_s) and these linking with other climatic, hydrological and water quality variables.

Considering assumptions in the modelling are the two types of the population under consideration, those that carry resistance genes, resistant bacteria (R), and those that do not carry resistance genes, susceptible bacteria (S) (Fig. 4). It means that the state of the population regarding these two classes of bacteria at a given moment t is determined by two population numbers, i.e., $R(t)$ and $S(t)$. Hence, the total population at any given time is:

$$N(t) = R(t) + S(t) \quad (2)$$

The means of change in the ARB population should be considered are: ARB population may increase because of vertical transmission of ARGs (growth or multiplication of bacteria), ARG could be acquired by susceptible bacteria through HGT, and mutations can select for the resistant mutants in several generations when the continuous presence of large antibiotic concentrations in the environment; ARB population may decrease because of death/decay and segregation or un-mutation (happens randomly in which resistant bacterium loses its resistance gene and becomes susceptible). The means of change in the susceptible bacteria population should be considered are: susceptible bacteria population may increase due to multiplication or growth and segregation or un-mutation; susceptible bacteria population may decrease because of converts into ARB through HGT, death/decay and mutations (susceptible bacterium converts to ARB). Therefore, susceptible bacteria become resistant either by acquiring ARGs via HGT mechanisms or through mutations.

The HGT via conjugation happens when susceptible and ARB physically interact with each other. Hence, the intensity of contact between these two is assumed to be proportional " $S \cdot R / (S+R)$ ", and the proportionality constant is given by β_c . HGT via transformation is when susceptible bacteria take up resistance genes floating in the nearby environment. Hence, this depends on the ARG numbers in the aquatic environment, which depends again on the lysis rate of the dead ARB and the shedding of their resistance gene in water. HGT via transformation occurs when a vector takes up the resistant bacteria's resistance gene and infects the susceptible bacteria. There is little information available for this kind of HGT. Hence, we need to assume that it is proportional to the concentration of resistant and susceptible bacteria with a proportionality constant of β_d . The following are the ordinary differential equations to represent the resistance gene dynamics in the aquatic environment:

$$\frac{dR}{dt} = (\mu_R \cdot R) - (D_R \cdot R) + \left(\beta_c \cdot \frac{S \cdot R}{S + R} + \beta_f \cdot S \cdot ARG + \beta_d \cdot S \cdot R \right) - \alpha \cdot S - \gamma \cdot R \quad (3)$$

$$\frac{DS}{Dt} = (\mu_S \cdot S) - (D_S \cdot S) - \left(\beta_c \cdot \frac{S \cdot R}{S + R} + \beta_f \cdot S \cdot ARG + \beta_d \cdot S \cdot R \right) - \alpha \cdot S + \gamma \cdot R \quad (4)$$

$$\frac{D(ARG)}{Dt} = l_{ysis} \cdot rate \cdot (D_S \cdot R) - \beta_f \cdot S \cdot (ARG) \quad (5)$$

The impact of chemical factors has been considered in terms of the antibiotics, heavy metal concentrations, DO and availability of nutrients and DOM. Monod's kinetic equation involves DOM concentration and can be employed in the model for simulating microbial growth. The heavy metals need to be assumed as a co-substrate which may inhibit the growth of susceptible bacteria. The AR mechanisms are usually associated with the fitness cost. It is known as the cost of carrying ARG, which is levied on the growth rate of ARB.

Overall, gene transport in the watershed can be expressed as follows in the modelling:

$$gene_m = gene_{m, i-1} * \exp(\mu_{net}) - gene_{dl} \quad (6)$$

$$gene_{ws} = gene_{water, i} + gene_r + gene_{vs} - gene_f - gene_s \pm gene_m \quad (7)$$

Where $gene_m$ is the number of copies present in all mediums of the watershed in a cubic meter area; i is the day; μ_{net} is the difference between growth and decay; $gene_{dl}$ is the daily loss of gene copies; $gene_{ws}$ is the number of copies in the total watershed; $gene_{water, i}$ is the number of gene copies from initial model step; $gene_r$ is resuspension from sediments or streambeds; $gene_{vs}$ is the genes released from vegetative filter strips around waterbodies; $gene_f$ is gene transport in water flow; $gene_s$ is the settled number of genes from the water body [149,161,125,202, 203]. Overall, this section explains a plausible model module considering several presumed variables for quantifying ARGs in a broader AR water quality modelling framework [31] (Fig. 4).

3.5. Model complexity control on predictions

The reliability of model predictions depends on bacterial population densities, growth and death, gene transfer rates, antibiotic and other pollutant concentrations, potentially antibiotic cycling and the influence of wider water quality conditions (DO, nutrients, etc.). Using different classes of antibiotics to reduce the development of resistance is termed antibiotic cycling. The resilience of antibiotic cycling comes into question when there are MDR bacteria in the environment, and the aim is to reduce the bacteria in the cycle. The interactions of antibiotic drug cocktails and ARB in the aquatic environments raise the question, how can alternative antibiotic classes influence the removal of microbes that have already become resistant over time? ARGs can endure in the environment long after removing the selection pressures, making antibiotic cycling ineffective [15]. It was even argued through a modelling study in hospital settings that antibiotic cycling can be unlikely to be effective in the long run in reducing or minimising resistance [204]. One should be careful to ascertain how any specific ARB can react to antibiotic cycling in the modelling and model predictions. This cannot be directly covered in initial fate and transport models considering the modelling options but need to look at the transport and fate of individual ARB or groups that act similarly.

All the model complexity can also perhaps be reduced to a simplistic model, which is how modelling should begin, and the complexity is only added to the model as it is needed to reduce model errors. But simplistic models must face greater uncertainties considering the nexus of antibiotics-ARB-ARGs. Many other model uncertainties and complexities might control the model predictions with respect to the model setting and geographical context:

- Problems with model complexity on the number of parameters come along when they are required to describe the dynamics and kinetics of gene transfer and propagation.
- Efficient modelling might require large sets of data in microcosm, field mesocosm and field studies in controlled and natural environments, which could pose an enormous challenge in envisaging AR conditions.
- Besides, model predictions carried out with higher uncertainties may not be helpful for pollution mitigation or management strategies but still valuable in designing experiments to understand the AR patterns in natural environment settings.

4. Priorities for development of efficient models

One of the key tasks for AR modelling is to identify the ARB populations and their link to the specific antibiotics and environments in which they propagate. New model development is necessary to address these complexities of AR. Specifically, to understand the interactions between antibiotics, ARB and ARGs and be able to respond to these questions:

- How can high concentrations of antibiotic residues (valid for LMICs) either drive resistance development or select for high levels of ARB (resistance already present)?
- How can ARG levels increase in a bacterial population and the ARG transfer rates from one bacterium to another?
- How can the microbiome composition change (affecting microbial structures) and act as a system stressor, assuming bacteria are in planktonic form and not sessile?
- How can nutrient availability in the environment support the growth of bacteria and aids their survival also influence these interactions?

Modelling a resistant bacterial population in a terrestrial water body will help us to identify the governing principles and dominant controlling processes. This should be based on regular spatio-temporal observations, as the ARB behaviour might be different from that of populations of the same family of pathogenic bacteria that are not resistant to antibiotics. Modelling should also enable us to understand ARB and ARGs' historical and future transport, which can help track down the sources and control human exposure to resistant pathogens. Finally, efficient modelling predictions can enable us to evaluate the effects of an intervention or plan interventions to prevent risks.

There is a trade-off between complex, process-based models and more simple 'black-box' mathematical relationships. The entry points and developments of these are limited either by our knowledge of the process or the absence of data, all of which can aggregate and create uncertainties. Nowhere is this more true than in the developing field of AR. The resistance development mechanisms, bacterial population growth rates and gene transfer mechanisms are complex processes to include in a water quality model. Issues and factors detailed in this article should be carefully considered in water quality modelling of ARB and ARGs, whether this involves modifying appropriate existing models, combining several existing models, or developing new models specifically to deal with AR issues. With any of these options, modelling can be quite tricky as there is no one-stop solution to model all the processes that control AR in different aquatic environments, given both our current level of process understanding of sources, transport and fate, as well as the data to characterise each of these elements. The foremost research priorities that should be considered for modelling are tabulated in Table 5. An element of developing ARB and ARGs fate and transport models should link them to epidemiological models based on the susceptible, infected and removed (SIR) paradigm to understand the interactions of resistant pathogens, environment and human health [205].

One of the significant issues faced with modelling is data availability and the quality of the data that can be obtained, as many countries around the world do not include ARB and ARGs in their regular water

Table 5

Foremost research priorities that should be considered in developing water quality models for evaluating the fate and transport processes of ARB.

Modelling steps and key factors	Remarks and priorities for modelling
Model components and boundaries	Identifying and determining the model components and boundary conditions to properly assess or analyse the inflows and outflows of ARB and ARG loads in the system boundary
Model setup	Designing modelling methodology or defining model objectives, or developing a modelling framework that can help to tackle the AR problem (one may not need to model all the complex processes in the system)
Collecting the essential and necessary AR data	Defining basic input parameters in the model should be considered essential, which includes minimal data on specific modelling environmental bacteria or resistant bacteria and ARGs information
ARB and ARGs source tracking	Ascertaining the sources of ARB and ARGs of the receiving waters in the modelling boundary can help determine the fate and transport conditions of the pathogen
Model capability to cope with limited data	Collecting antibiotic resistance and chemical concentration data in aquatic environments is often costly, especially in LMICs. Using proxy data of antibiotic resistance and resistance determinants might be useful but have substantial limitations
Model calibration and validation	Model complexity, uncertainties, limited data and antibiotic resistance determinants certainly will influence the calibration and validation of the fate and transport models
Antibiotic loadings to water systems	Information on antibiotic loadings shows both the source and quantity of antibiotics necessary to determine breeding environments and calculate ARB and ARG concentrations in the water environments. Large concentrations of antibiotics in the water environments may determine the control of selection pressure on resistant bacteria's fate and transport
Intricate implications of ARGs	Any information on ARGs is critical in fate and transport modelling because it may strongly influence the environmental bacteria to become resistant or resistant bacteria to acquire multidrug resistance
Nexus of Antibiotics-ARB-ARGs	The complexity starts when the bacteria interact with antibiotics and resistance genes but varies depending on the medium, and the resistant bacteria's survival or growth depends on entangled links among these three water quality variables
Multidrug-resistant (MDR) bacteria	It is probably tough to recognise, determine, or predict the bacteria that acquire or lose multidrug resistance in modelling. But it is imperative to understand the multidrug resistance in the water environments using fate and transport modelling
The influence of organic and inorganic pollutant loadings	In no other controlled or natural environments resistant bacteria will have a longer and cumulative exposure than in aquatic ecosystems. In ARB fate and transport modelling, one should look for the impact of pH, salinity, nutrient availability, heavy metal resistance, pesticides, oxygen conditions, etc.

quality monitoring programs. Therefore, it is necessary for models to cope with limited AR data, especially in LMICs (further, please see supplementary section S5). Also, we need to conduct more microcosm experiments studying the impact of various environmental variables like temperature, pH, salinity, nutrient concentration, and oxygen saturation on the acquisition and propagation of AR in various faecal and pathogenic bacteria. This can improve the development of fate and transport models for mesocosm and field-scale studies. Again, there might be different starting points depending on the modelling objective. Finally, researchers and modellers must look for the resistance evolving patterns in ESKAPE bacteria (*Enterococcus faecium*, *Staphylococcus aureus*, *Klebsiella pneumoniae*, *Acinetobacter baumannii*, *Pseudomonas aeruginosa*, and *Enterobacter spp.*), in which multidrug resistance has been rapidly evolving in recent years.

5. Conclusions and recommendations

ARB and ARGs are increasingly reported in various aquatic environments, and understanding their fate and transport processes is limited. Modelling is an essential tool for analysing AR in water systems that can help us understand spatial and temporal variations of ARB and ARGs, which can be critical in developing pollution management strategies. Moreover, AR development can give bacteria greater environmental fitness and provide a comparative advantage that may select for ARB dominance in the environment. Therefore, it is essential to understand and assimilate how environmental fitness translates into higher survival and/or lower decay rates in ARB. Sophisticated water quality models seek to analyse how complex hydrological, climatic, and ecological factors and processes influence the fate and transport of (non-resistant) bacteria in waters, which could be translated to model ARB. However, existing water quality models will need to be adapted and complemented to incorporate the processes that govern resistance development in water systems, including gene transfer and selective pressures. Furthermore, it is crucial to incorporate information on novel ARGs and other resistance development mechanisms in different aquatic environments. It is also important to understand and incorporate in the modelling how different contaminants (e.g., heavy metals, pharmaceuticals, pesticides, etc.) interact with waterborne pathogenic bacteria and how they can influence AR development and bacterial behaviour. Data

availability will also be critical to calibrate and validate water quality models. Local planners and decision-makers should improve the surveillance of ARB and ARGs on local water bodies. Integrating different model modules is key for understanding ARB and ARG fate, transport, exposure and associated health risks. Such models would also allow analysing, ex-ante, the effectiveness of different risk mitigation options, including improved Water, Sanitation and Hygiene (WASH). New fate and transport models and their predictive capability can bestow sustainable measures and practices to control the spreading of AR to aquatic ecosystems and help ensure healthy lives and well-being.

Environmental implication

Antibiotic resistance (AR) is promoted by the imprudent use of antimicrobial agents across all sectors and the release of resultant wastes and other pollutants into the natural environment. Consequently, animal and human exposure to resistant microorganisms are increasing, including pathogens, especially via waterborne pathways. Despite the potential importance of waterborne AR as a health risk, only a few reliable numerical models have been developed that simulate AR dynamics in water systems. This review provides insights into water quality modelling to guide model development of the fate and transport of AR bacteria and genes - models critical for performing future risk assessments on AR spread and exposure in water systems.

Glossary

Antibiotics	Medicines that treat bacterial infections by preventing the growth or killing of bacteria
Antibiotic resistance (AR)	When bacteria develop resistance and can survive exposure to antibiotics
Antibiotic resistance genes (ARGs)	Genes that encode mechanisms that confer resistance to antibiotics
Antibiotic resistant bacteria (ARB)	Bacteria that have resistance mechanisms and can survive exposure to antibiotics
Bacteriophages	Viruses that infect and replicate within bacteria
Mobile Genetic Elements (MGEs)	Transferable determinants that can harbour antibiotic resistance genes, e.g. transposons, integrons, plasmids.
Environmental fitness	The ability of bacteria to survive in harsh environmental conditions

(continued on next page)

(continued)

Microbiome	Composition or structure of the microbial population in a particular ecological niche
Multidrug-resistant bacteria	Bacteria that exhibit resistance to more than one antibiotic class
Receiving waters	River, lake, or any watercourse that receives wastewater or treated effluent
Resistome	Total antibiotic resistance gene content or resistance genes assemblage
Virome	Total composition of viruses or virus assemblage

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This study is supported by the Foreign, Commonwealth and Development Office (FCDO, UK) grant to the CGIAR research program on Agriculture for Nutrition and Health (A4NH) led by IFPRI and implemented through the CGIAR Antimicrobial Resistance Hub, CGIAR research program on Fish Agri-Food Systems led by WorldFish, CGIAR research program on Water, Land and Ecosystems (WLE) led by IWMI, and the CGIAR Initiative on One Health (Protecting human health through a One Health approach). We are thankful to Prof. Jeff Waage of the London School of Hygiene & Tropical Medicine (LSHTM) and Mr Terry Clayton for their constructive comments and suggestions to improve the quality of the paper.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.jhazmat.2023.132527](https://doi.org/10.1016/j.jhazmat.2023.132527).

References

- [1] Hutchings, M.I., Truman, A.W., Wilkinson, B., 2019. Antibiotics: past, present and future. *Curr Opin Microbiol*, Antimicrob 51, 72–80. <https://doi.org/10.1016/j.mib.2019.10.008>.
- [2] Davies, J., Davies, D., 2010. Origins and evolution of antibiotic resistance. *Microbiol Mol Biol Rev* 74, 417–433. <https://doi.org/10.1128/MMBR.00016-10>.
- [3] Martínez, J.L., 2008. Antibiotics and antibiotic resistance genes in natural environments. *Science* 321, 365–367. <https://doi.org/10.1126/science.1159483>.
- [4] WHO, 2017. Global antimicrobial resistance surveillance system (GLASS) report: early implementation 2016–2017. World Health Organization, Geneva.
- [5] Jampani, M., Chandy, S.J., 2021. Increased antimicrobial use during COVID-19: the risk of advancing the threat of antimicrobial resistance. *Health Sci Rep* 4, e459. <https://doi.org/10.1002/hsr.2.459>.
- [6] Lupo, A., Coyne, S., Berendonk, T.U., 2012. Origin and evolution of antibiotic resistance: the common mechanisms of emergence and spread in water bodies. *Front Microbiol* 3. <https://doi.org/10.3389/fmicb.2012.00018>.
- [7] O'Neill, J., 2016. Tackling Drug-Resistant Infections Globally: final report and recommendations. The Review on Antimicrobial Resistance. Wellcome Trust.
- [8] Murray, C.J., Ikuta, K.S., Sharara, F., Swetschinski, L., Robles Aguilar, G., Gray, A., Han, C., Bisignano, C., Rao, P., Wool, E., Johnson, S.C., Browne, A.J., Chipeta, M.G., Fell, F., Hackett, S., Haines-Woodhouse, G., Kashef Hamadani, B. H., Kumaran, E.A.P., McManigal, B., Agarwal, R., Akech, S., Albertson, S., Amuasi, J., Andrews, J., Aravkin, A., Ashley, E., Bailey, F., Baker, S., Basnyat, B., Bekker, A., Bender, R., Bethou, A., Bielicki, J., Boonkasidecha, S., Bukosia, J., Carvalho, C., Castañeda-Orjuela, C., Chansamouth, V., Chaurasia, S., Chiurichiù, S., Chowdhury, F., Cook, A.J., Cooper, B., Cressey, T.R., Criollo-Mora, E., Cunningham, M., Darboe, S., Day, N.P.J., De Luca, M., Dokova, K., Dramowski, A., Dunachie, S.J., Eckmanns, T., Eibach, D., Emami, A., Feasey, N., Fisher-Pearson, N., Forrest, K., Garrett, D., Gastmeier, P., Giref, A.Z., Greer, R.C., Gupta, V., Haller, S., Haselbeck, A., Hay, S.I., Holm, M., Hopkins, S., Iregbu, K.C., Jacobs, J., Jarovsky, D., Javanmardi, F., Khorana, M., Kissoon, N., Kobeissi, E., Kostyaney, T., Krapp, F., Krumkamp, R., Kumar, A., Kyu, H.H., Lim, C., Limmathurotsakul, D., Loftus, M.J., Lunn, M., Ma, J., Mturi, N., Munera-Huertas, T., Musicha, P., Mussi-Pinhata, M.M., Nakamura, T., Nanavati, R., Nangia, S., Newton, P., Ngoun, C., Novotney, A., Nwakanma, D., Obiero, C.W., Olivas-Martinez, A., Olliaro, P., Ooko, E., Ortiz-Brizuela, E., Peleg, A.Y., Perrone, C., Plakkal, N., Ponce-de-Leon, A., Raad, M., Ramdin, T., Riddell, A., Roberts, T., Robotham, J.V., Roca, A., Rudd, K.E., Russell, N., Schnall, J., Scott, J. A.G., Shivamallappa, M., Sifuentes-Osorio, J., Steenkeste, N., Stewardson, A.J., Stoeva, T., Tasak, N., Thaiprakong, A., Thwaites, G., Turner, C., Turner, P., van Doorn, H.R., Velaphi, S., Vongpradith, A., Vu, H., Walsh, T., Waner, S., Wangrangsamakul, T., Wozniak, T., Zheng, P., Sartorius, B., Lopez, A.D., Stergachis, A., Moore, C., Dolecek, C., Naghavi, M., 2022. Global burden of bacterial antimicrobial resistance in 2019: a systematic analysis. *Lancet*. [https://doi.org/10.1016/S0140-6736\(21\)02724-0](https://doi.org/10.1016/S0140-6736(21)02724-0).
- [9] World Bank, 2017. Drug-Resistant Infections: A Threat to Our Economic Future. World Bank, Washington, DC.
- [10] World Economic Forum, 2021. Antimicrobial Resistance and Water: The Risks and Costs for Economies and Societies (Briefing Paper).
- [11] Brunn, A., Kadri-Alabi, Z., Moodley, A., Guardabassi, L., Taylor, P., Mateus, A., Waage, J., 2022. Characteristics and global occurrence of human pathogens harboring antimicrobial resistance in food crops: a scoping review. *Front Sustain Food Syst* 6.
- [12] Mulchandani, R., Wang, Y., Gilbert, M., Boeckel, T.P.V., 2023. Global trends in antimicrobial use in food-producing animals: 2020 to 2030. *PLOS Glob Public Health* 3, e0001305. <https://doi.org/10.1371/journal.pgph.0001305>.
- [13] Zhang, Z., Zhang, Q., Wang, T., Xu, N., Lu, T., Hong, W., Penuelas, J., Gillings, M., Wang, M., Gao, W., Qian, H., 2022. Assessment of global health risk of antibiotic resistance genes. *Nat Commun* 13, 1553. <https://doi.org/10.1038/s41467-022-29283-8>.
- [14] World Bank, 2019. Pulling Together to Beat Superbugs: Knowledge and Implementation Gaps in Addressing Antimicrobial Resistance. World Bank, Washington, DC.
- [15] Taylor, N.G.H., Verner-Jeffreys, D.W., Baker-Austin, C., 2011. Aquatic systems: maintaining, mixing and mobilising antimicrobial resistance? *Trends Ecol Evol* 26, 278–284. <https://doi.org/10.1016/j.tree.2011.03.004>.
- [16] Frost, L.S., Leplae, R., Summers, A.O., Toussaint, A., 2005. Mobile genetic elements: the agents of open source evolution. *Nat Rev Microbiol* 3, 722–732. <https://doi.org/10.1038/nrmicro1235>.
- [17] Partridge, S.R., Kwong, S.M., Firth, N., Jensen, S.O., 2018. Mobile genetic elements associated with antimicrobial resistance. *Clin Microbiol Rev* 31. <https://doi.org/10.1128/CMR.00088-17>.
- [18] Andersson, D.I., Hughes, D., 2014. Microbiological effects of sublethal levels of antibiotics. *Nat Rev Microbiol* 12, 465–478. <https://doi.org/10.1038/nrmicro3270>.
- [19] Holmes, A.H., Moore, L.S.P., Sundsfjord, A., Steinbakk, M., Regmi, S., Karkey, A., Guerin, P.J., Piddock, L.J.V., 2016. Understanding the mechanisms and drivers of antimicrobial resistance. *Lancet* 387, 176–187. [https://doi.org/10.1016/S0140-6736\(15\)00473-0](https://doi.org/10.1016/S0140-6736(15)00473-0).
- [20] Marti, E., Variatza, E., Balcasar, J.L., 2014. The role of aquatic ecosystems as reservoirs of antibiotic resistance. *Trends Microbiol* 22, 36–41. <https://doi.org/10.1016/j.tim.2013.11.001>.
- [21] Martínez, J.L., Coque, T.M., Baquero, F., 2015. What is a resistance gene? Ranking risk in resistomes. *Nat Rev Microbiol* 13, 116–123. <https://doi.org/10.1038/nrmicro3399>.
- [22] Munita, J.M., Arias, C.A., 2016. Mechanisms of Antibiotic Resistance. in: *Virulence Mechanisms of Bacterial Pathogens*. John Wiley & Sons, Ltd., pp. 481–511. <https://doi.org/10.1128/9781555819286.ch17>.
- [23] Sultan, I., Rahman, S., Jan, A.T., Siddiqui, M.T., Mondal, A.H., Haq, Q.M.R., 2018. Antibiotics, resistome and resistance mechanisms: a bacterial perspective. *Front Microbiol* 9. <https://doi.org/10.3389/fmicb.2018.02066>.
- [24] Tenover, F.C., 2006. Mechanisms of antimicrobial resistance in bacteria. *Am J Med*, Antimicrob Resist Prev Initiat: Proc Expert Panel Resist 119, S3–S10. <https://doi.org/10.1016/j.ajmed.2006.03.011>.
- [25] Larsson, D.G.J., Flach, C.-F., 2022. Antibiotic resistance in the environment. *Nat Rev Microbiol* 20, 257–269. <https://doi.org/10.1038/s41579-021-00649-x>.
- [26] UNEP, 2023. Bracing for Superbugs: Strengthening environmental action in the One Health response to antimicrobial resistance. Geneva.
- [27] Bengtsson-Palme, J., Kristiansson, E., Larsson, D.G.J., 2018. Environmental factors influencing the development and spread of antibiotic resistance. *FEMS Microbiol Rev* 42. <https://doi.org/10.1093/femsrev/fux053>.
- [28] Chereau, F., Opatowski, L., Tourdjman, M., Vong, S., 2017. Risk assessment for antibiotic resistance in South East Asia. *BMJ* 358. <https://doi.org/10.1136/bmj.j3393>.
- [29] Ashbolt, N.J., Amézquita, A., Backhaus, T., Borriello, P., Brandt, K.K., Collignon, P., Coors, A., Finley, R., Gaze, W.H., Heberer, T., Lawrence, J.R., Larsson, D.G.J., McEwen, S.A., Ryan, J.J., Schönfeld, J., Silley, P., Snape, J.R., Van den Eede, C., Topp, E., 2013. Human Health Risk Assessment (HHRA) for Environmental Development and Transfer of Antibiotic Resistance. *Environ Health Perspect* 121, 993–1001. <https://doi.org/10.1289/ehp.1206316>.
- [30] UNEP, 2022. Environmental Dimensions of Antimicrobial Resistance: Summary for Policymakers. United Nations Environment Programme.
- [31] Jampani, M., Gothwal, R., Mateo-Sagasta, J., Langan, S., 2022. Water quality modelling framework for evaluating antibiotic resistance in aquatic environments. *J Hazard Mater Lett* 3, 100056. <https://doi.org/10.1016/j.hazl.2022.100056>.
- [32] Graham, D.W., Bergeron, G., Bourassa, M.W., Dickson, J., Gomes, F., Howe, A., Kahn, L.H., Morley, P.S., Scott, H.M., Simjee, S., Singer, R.S., Smith, T.C., Storrs, C., Wittum, T.E., 2019. Complexities in understanding antimicrobial resistance across domesticated animal, human, and environmental systems. *Ann N Y Acad Sci* 1441, 17–30. <https://doi.org/10.1111/nyas.14036>.
- [33] Wellington, E.M., Boxall, A.B., Cross, P., Feil, E.J., Gaze, W.H., Hawkey, P.M., Johnson-Rollings, A.S., Jones, D.L., Lee, N.M., Otten, W., Thomas, C.M.,

- Williams, A.P., 2013. The role of the natural environment in the emergence of antibiotic resistance in Gram-negative bacteria. *Lancet Infect Dis* 13, 155–165. [https://doi.org/10.1016/S1473-3099\(12\)70317-1](https://doi.org/10.1016/S1473-3099(12)70317-1).
- [34] Taneja, N., Sharma, M., 2019. Antimicrobial resistance in the environment: the Indian scenario. *Indian J Med Res* 149, 119–128. https://doi.org/10.4103/ijmr.IJMR_331_18.
- [35] Perianes-Rodriguez, A., Waltman, L., van Eck, N.J., 2016. Constructing bibliometric networks: a comparison between full and fractional counting. *J Informetr* 10, 1178–1195. <https://doi.org/10.1016/j.joi.2016.10.006>.
- [36] Bielen, A., Šimatović, A., Kosić-Vukšić, J., Senta, I., Ahel, M., Babić, S., Jurina, T., González Plaza, J.J., Milaković, M., Udiković-Kolić, N., 2017. Negative environmental impacts of antibiotic-contaminated effluents from pharmaceutical industries. *Water Res* 126, 79–87. <https://doi.org/10.1016/j.watres.2017.09.019>.
- [37] Boeckel, T.P.V., Pires, J., Silvester, J., Zhao, C., Song, J., Criscuolo, N.G., Gilbert, M., Bonhoeffer, S., Laxminarayan, R., 2019. Global trends in antimicrobial resistance in animals in low- and middle-income countries. *Science* 365. <https://doi.org/10.1126/science.aaw1944>.
- [38] Cauci, S., Karkman, A., Cacace, D., Rybicki, M., Timpel, P., Voilaid, V., Gurke, R., Virta, M., Berendonk, T.U., 2016. Seasonality of antibiotic prescriptions for outpatients and resistance genes in sewers and wastewater treatment plant outflow. *FEMS Microbiol Ecol* 92. <https://doi.org/10.1093/femsec/fiw060>.
- [39] Hassoun-Kheir, N., Stabholz, Y., Kreft, J.-U., de la Cruz, R., Romalde, J.L., Nesme, J., Sørensen, S.J., Smets, B.F., Graham, D., Paul, M., 2020. Comparison of antibiotic-resistant bacteria and antibiotic resistance genes abundance in hospital and community wastewater: A systematic review. *Sci Total Environ* 743, 140804. <https://doi.org/10.1016/j.scitotenv.2020.140804>.
- [40] Pärnänen, K.M.M., Narciso-da-Rocha, C., Kneis, D., Berendonk, T.U., Cacace, D., Do, T.T., Elpers, C., Fatta-Kassinos, D., Henriques, I., Jaeger, T., Karkman, A., Martínez, J.L., Michael, S.G., Michael-Kordatou, I., O'Sullivan, K., Rodríguez-Mozaz, S., Schwartz, T., Sheng, H., Sørum, H., Stedtfeld, R.D., Tiedje, J.M., Giustina, S.V.D., Walsh, F., Vaz-Moreira, I., Virta, M., Manaia, C.M., 2019. Antibiotic resistance in European wastewater treatment plants mirrors the pattern of clinical antibiotic resistance prevalence. *Sci Adv* 5, eaau9124. <https://doi.org/10.1126/sciadv.aau9124>.
- [41] Karkman, A., Pärnänen, K., Larsson, D.G.J., 2019. Fecal pollution can explain antibiotic resistance gene abundances in anthropogenically impacted environments. *Nat Commun* 10 (1), 8. <https://doi.org/10.1038/s41467-018-07992-3>.
- [42] Le, T.-H., Ng, C., Chen, H., Yi, X.Z., Koh, T.H., Barkham, T.M.S., Zhou, Z., Gin, K. Y.-H., 2016. Occurrences and characterization of antibiotic-resistant bacteria and genetic determinants of hospital wastewater in a tropical country. *Antimicrob Agents Chemother* 60, 7449–7456. <https://doi.org/10.1128/AAC.01556-16>.
- [43] Corno, G., Yang, Y., Eckert, E.M., Fontaneto, D., Fiorentino, A., Galafassi, S., Zhang, T., Di Cesare, A., 2019. Effluents of wastewater treatment plants promote the rapid stabilization of the antibiotic resistome in receiving freshwater bodies. *Water Res* 158, 72–81. <https://doi.org/10.1016/j.watres.2019.04.031>.
- [44] DOH, 2014. Antimicrobial Resistance (AMR) Systems Map: Overview of the factors influencing the development of AMR and the interactions between them. Department of Health, London.
- [45] Wright, G.D., 2019. Environmental and clinical antibiotic resistomes, same only different. *Curr Opin Microbiol, Antimicrob* 51, 57–63. <https://doi.org/10.1016/j.mib.2019.06.005>.
- [46] Saima, S., Fiaz, M., Zafar, R., Ahmed, I., Arshad, M., 2020. Dissemination of antibiotic resistance in the environment. in: *Antibiotics and Antimicrobial Resistance Genes in the Environment*. Elsevier, pp. 99–116. <https://doi.org/10.1016/B978-0-12-818882-8.00006-1>.
- [47] Yang, Y., Song, W., Lin, H., Wang, W., Du, L., Xing, W., 2018. Antibiotics and antibiotic resistance genes in global lakes: a review and meta-analysis. *Environ Int* 116, 60–73. <https://doi.org/10.1016/j.envint.2018.04.011>.
- [48] Uluseker, C., Kaster, K.M., Thorsen, K., Basiry, D., Shobana, S., Jain, M., Kumar, G., Kommedal, R., Pala-Ozkok, I., 2021. A Review on Occurrence and Spread of Antibiotic Resistance in Wastewaters and in Wastewater Treatment Plants: Mechanisms and Perspectives. *Front Microbiol* 12.
- [49] Garcia-Armisen, T., Anzil, A., Cornelis, P., Chevreuil, M., Servais, P., 2013. Identification of antimicrobial resistant bacteria in rivers: Insights into the cultivation bias. *Water Res* 47, 4938–4947. <https://doi.org/10.1016/j.watres.2013.05.036>.
- [50] Kittinger, C., Lipp, M., Folli, B., Kirschner, A., Baumert, R., Galler, H., Grisold, A. J., Luxner, J., Weissenbacher, M., Farnleitner, A.H., Zarfel, G., 2016. Enterobacteriaceae Isolated from the River Danube: Antibiotic Resistances, with a Focus on the Presence of ESBL and Carbapenemases. *PLOS ONE* 11, e0165820. <https://doi.org/10.1371/journal.pone.0165820>.
- [51] Martí, E., Jofre, J., Balcázar, J.L., 2013. Prevalence of antibiotic resistance genes and bacterial community composition in a river influenced by a wastewater treatment plant. *PLOS ONE* 8, e78906. <https://doi.org/10.1371/journal.pone.0078906>.
- [52] Cesare, A.D., Eckert, E.M., Teruggi, A., Fontaneto, D., Bertoni, R., Callieri, C., Corno, G., 2015. Constitutive presence of antibiotic resistance genes within the bacterial community of a large subalpine lake. *Mol Ecol* 24, 3888–3900. <https://doi.org/10.1111/mec.13293>.
- [53] Huerta, B., Martí, E., Gros, M., López, P., Pompéo, M., Armengol, J., Barceló, D., Balcázar, J.L., Rodríguez-Mozaz, S., Marcé, R., 2013. Exploring the links between antibiotic occurrence, antibiotic resistance, and bacterial communities in water supply reservoirs, 161–170 *Sci Total Environ* 456–457. <https://doi.org/10.1016/j.scitotenv.2013.03.071>.
- [54] Carney, R.L., Labbate, M., Siboni, N., Tagg, K.A., Mitrovic, S.M., Seymour, J.R., 2019. Urban beaches are environmental hotspots for antibiotic resistance following rainfall. *Water Res* 167, 115081. <https://doi.org/10.1016/j.watres.2019.115081>.
- [55] Chee-Sanford, J.C., Aminov, R.I., Krapac, I.J., Garrigues-Jeanjean, N., Mackie, R. I., 2001. Occurrence and diversity of tetracycline resistance genes in lagoons and groundwater underlying two swine production facilities. *Appl Environ Microbiol* 67, 1494–1502. <https://doi.org/10.1128/AEM.67.4.1494-1502.2001>.
- [56] Andrade, L., Kelly, M., Hynds, P., Weatherill, J., Majury, A., O'Dwyer, J., 2020. Groundwater resources as a global reservoir for antimicrobial-resistant bacteria. *Water Res* 170, 115360. <https://doi.org/10.1016/j.watres.2019.115360>.
- [57] Cerqueira, F., Matamoros, V., Bayona, J., Elsinga, G., Hornstra, L.M., Piña, B., 2019. Distribution of antibiotic resistance genes in soils and crops. A field study in legume plants (*Vicia faba* L.) grown under different watering regimes. *Environ Res* 170, 16–25. <https://doi.org/10.1016/j.envres.2018.12.007>.
- [58] Ribeiro, A.F., Bodilis, J., Alonso, L., Buquet, S., Feuilloley, M., Dupont, J.-P., Pawlak, B., 2014. Occurrence of multi-antibiotic resistant *Pseudomonas* spp. in drinking water produced from karstic hydrosystems. *Sci Total Environ* 490, 370–378. <https://doi.org/10.1016/j.scitotenv.2014.05.012>.
- [59] UNEP, 2016. A Snapshot of the World's Water Quality: Towards a global assessment. United Nations Environment Programme, Nairobi, Kenya.
- [60] McConnell, M.M., Hansen, L.T., Neudorf, K.D., Hayward, J.L., Jamieson, R.C., Yost, C.K., Tong, A., 2018. Sources of Antibiotic Resistance Genes in a Rural River System. *J Environ Qual* 47, 997–1005. <https://doi.org/10.2134/jeq2017.12.0477>.
- [61] Larsson, D.G.J., Andremon, A., Bengtsson-Palme, J., Brandt, K.K., de Roda Husman, A.M., Fagerstedt, P., Fick, J., Flach, C.-F., Gaze, W.H., Kuroda, M., Kvint, K., Laxminarayan, R., Manaia, C.M., Nielsen, K.M., Plant, L., Ploy, M.-C., Segovia, C., Simonet, P., Smalla, K., Snape, J., Topp, E., van Hengel, A.J., Verner-Jeffreys, D.W., Virta, M.P.J., Wellington, E.M., Wernersson, A.-S., 2018. Critical knowledge gaps and research needs related to the environmental dimensions of antibiotic resistance. *Environ Int* 117, 132–138. <https://doi.org/10.1016/j.envint.2018.04.041>.
- [62] Chen, Y., Li, P., Huang, Y., Yu, K., Chen, H., Cui, K., Huang, Q., Zhang, J., Yew-Hoong Gin, K., He, Y., 2019. Environmental media exert a bottleneck in driving the dynamics of antibiotic resistance genes in modern aquatic environment. *Water Res* 162, 127–138. <https://doi.org/10.1016/j.watres.2019.06.047>.
- [63] Rizzo, L., Manaia, C., Merlin, C., Schwartz, T., Dagot, C., Ploy, M.C., Michael, I., Fatta-Kassinos, D., 2013. Urban wastewater treatment plants as hotspots for antibiotic resistant bacteria and genes spread into the environment: a review. *Sci Total Environ* 447, 345–360. <https://doi.org/10.1016/j.scitotenv.2013.01.032>.
- [64] Graham, D.W., Giesen, M.J., Bunce, J.T., 2019. Strategic Approach for Prioritising Local and Regional Sanitation Interventions for Reducing Global Antibiotic Resistance. *Water* 11, 27. <https://doi.org/10.3390/w11010027>.
- [65] Pei, R., Cha, J., Carlson, K.H., Pruden, A., 2007. Response of Antibiotic Resistance Genes (ARG) to Biological Treatment in Dairy Lagoon Water. *Environ Sci Technol* 41, 5108–5113. <https://doi.org/10.1021/es070051x>.
- [66] Qian, M., Yang, L., Chen, X., Li, K., Xue, W., Li, Y., Zhao, H., Cao, G., Guan, X., Shen, G., 2020. The treatment of veterinary antibiotics in swine wastewater by biodegradation and Fenton-like oxidation. *Sci Total Environ* 710, 136299. <https://doi.org/10.1016/j.scitotenv.2019.136299>.
- [67] Quintela-Baluja, M., Abouelnga, M., Romalde, J., Su, J.-Q., Yu, Y., Gomez-Lopez, M., Smets, B., Zhu, Y.-G., Graham, D.W., 2019. Spatial ecology of a wastewater network defines the antibiotic resistance genes in downstream receiving waters. *Water Res* 162, 347–357. <https://doi.org/10.1016/j.watres.2019.06.075>.
- [68] Christgen, B., Yang, Y., Ahammad, S.Z., Li, B., Rodriguez, D.C., Zhang, T., Graham, D.W., 2015. Metagenomics shows that low-energy anaerobic treatment reactors reduce antibiotic resistance gene levels from domestic wastewater. *Environ Sci Technol* 49, 2577–2584. <https://doi.org/10.1021/es505521w>.
- [69] Schwermer, C.U., Krzeminski, P., Wennberg, A.C., Vogelsang, C., Uhl, W., 2017. Removal of antibiotic resistant *E. coli* in two Norwegian wastewater treatment plants and by nano- and ultra-filtration processes. *Water Sci Technol* 77, 1115–1126. <https://doi.org/10.2166/wst.2017.642>.
- [70] Thwaites, B.J., Short, M.D., Stuetz, R.M., Reeve, P.J., Alvarez Gaitan, J.-P., Dinesh, N., van den Akker, B., 2018. Comparing the performance of aerobic granular sludge versus conventional activated sludge for microbial log removal and effluent quality: Implications for water reuse. *Water Res* 145, 442–452. <https://doi.org/10.1016/j.watres.2018.08.038>.
- [71] Baker, M., Hobman, J.L., Dodd, C.E.R., Ramsden, S.J., Stelkel, D.J., 2016. Mathematical modelling of antimicrobial resistance in agricultural waste highlights importance of gene transfer rate. *FEMS Microbiol Ecol* 92, fiw040. <https://doi.org/10.1093/femsec/fiw040>.
- [72] Indorica, S., Ching, C., Darash, D., Suprenant, M., Emma, B., Heins, Z., Khalil, A. S., Zaman, M.H., 2021. Computational model to quantify the growth of antibiotic-resistant bacteria in wastewater. *MSystems* 6. <https://doi.org/10.1128/mSystems.00360-21>.
- [73] Polesel, F., Andersen, H.R., Trapp, S., Plósz, B.Gy, 2016. Removal of antibiotics in biological wastewater treatment systems—a critical assessment using the activated sludge modeling framework for xenobiotics (ASM-X). *Environ Sci Technol* 50, 10316–10334. <https://doi.org/10.1021/acs.est.6b01899>.
- [74] D'Costa, V.M., King, C.E., Kalan, L., Morar, M., Sung, W.W.L., Schwarz, C., Froese, D., Zazula, G., Calmels, F., Debruyne, R., Golding, G.B., Poinar, H.N.,

- Wright, G.D., 2011. Antibiotic resistance is ancient. *Nature* 477, 457–461. <https://doi.org/10.1038/nature10388>.
- [75] Forsberg, K.J., Reyes, A., Wang, B., Selleck, E.M., Sommer, M.O.A., Dantas, G., 2012. The Shared Antibiotic Resistome of Soil Bacteria and Human Pathogens. *Science* 337, 1107–1111. <https://doi.org/10.1126/science.1220761>.
- [76] Jauregi, L., Epelde, L., Alkorta, I., Garbisu, C., 2021. Antibiotic resistance in agricultural soil and crops associated to the application of cow manure-derived amendments from conventional and organic livestock farms. *Front Vet Sci* 8.
- [77] Joy, S.R., Bartelt-Hunt, S.L., Snow, D.D., Gilley, J.E., Woodbury, B.L., Parker, D. B., Marx, D.B., Li, X., 2013. Fate and transport of antimicrobials and antimicrobial resistance genes in soil and runoff following land application of swine manure slurry. *Environ Sci Technol* 47, 12081–12088. <https://doi.org/10.1021/es4026358>.
- [78] Sun, J., Jin, L., He, T., Wei, Z., Liu, X., Zhu, L., Li, X., 2020. Antibiotic resistance genes (ARGs) in agricultural soils from the Yangtze River Delta, China. *Sci Total Environ* 740, 140001. <https://doi.org/10.1016/j.scitotenv.2020.140001>.
- [79] Zhang, J., Sui, Q., Tong, J., Zhong, H., Wang, Y., Chen, M., Wei, Y., 2018. Soil types influence the fate of antibiotic-resistant bacteria and antibiotic resistance genes following the land application of sludge composts. *Environ Int* 118, 34–43. <https://doi.org/10.1016/j.envint.2018.05.029>.
- [80] Graham, D.W., Knapp, C.W., Christensen, B.T., McCluskey, S., Dolfing, J., 2016. Appearance of β -lactam Resistance Genes in Agricultural Soils and Clinical Isolates over the 20th Century. *Sci Rep* 6, 21550. <https://doi.org/10.1038/srep21550>.
- [81] Burch, T., Sadowsky, M., LaPara, T., 2013. Aerobic digestion reduces the quantity of antibiotic resistance genes in residual municipal wastewater solids. *Front Microbiol* 4.
- [82] Fahrenfeld, N., Ma, Y., O'Brien, M., Pruden, A., 2013. Reclaimed water as a reservoir of antibiotic resistance genes: distribution system and irrigation implications. *Front Microbiol* 4, 130. <https://doi.org/10.3389/fmicb.2013.00130>.
- [83] Yang, H., Byelashov, O.A., Geomaras, I., Goodridge, L.D., Nightingale, K.K., Belk, K.E., Smith, G.C., Sofos, J.N., 2010. Presence of antibiotic-resistant commensal bacteria in samples from agricultural, city, and national park environments evaluated by standard culture and real-time PCR methods. *Can J Microbiol* 56, 761–770. <https://doi.org/10.1139/W10-060>.
- [84] Amin, M.G.M., Forslund, A., Bui, X.T., Juhler, R.K., Petersen, S.O., Lægdsmand, M., 2013. Persistence and leaching potential of microorganisms and mineral N in animal manure applied to intact soil columns. *Appl Environ Microbiol* 79, 535–542. <https://doi.org/10.1128/AEM.02506-12>.
- [85] Bai, H., Cochet, N., Pauss, A., Lamy, E., 2016. Bacteria cell properties and grain size impact on bacteria transport and deposition in porous media. *Colloids Surf B Biointerfaces* 139, 148–155. <https://doi.org/10.1016/j.colsurfb.2015.12.016>.
- [86] Bolster, C.H., Brooks, J.P., Cook, K.L., 2018. Effect of manure application rate and rainfall timing on the leaching of antibiotic-resistant bacteria and their associated genes. *Water Air Soil Pollut* 229, 130. <https://doi.org/10.1007/s11270-018-3781-6>.
- [87] Garder, J.L., Moorman, T.B., Soupir, M.L., 2014. Transport and Persistence of Tylosin-Resistant Enterococci, erm Genes, and Tylosin in Soil and Drainage Water from Fields Receiving Swine Manure. *J Environ Qual* 43, 1484–1493. <https://doi.org/10.2134/jeq2013.09.0379>.
- [88] Poté, J., Ceccherini, M.T., Van, V.T., Rosselli, W., Wildi, W., Simonet, P., Vogel, T. M., 2003. Fate and transport of antibiotic resistance genes in saturated soil columns. *Eur J Soil Biol* 39, 65–71. [https://doi.org/10.1016/S1164-5563\(03\)00003-7](https://doi.org/10.1016/S1164-5563(03)00003-7).
- [89] Barrios, R.E., Bartelt-Hunt, S.L., Li, Y., Li, X., 2021. Modeling the vertical transport of antibiotic resistance genes in agricultural soils following manure application. *Environ Pollut* 285, 117480. <https://doi.org/10.1016/j.envpol.2021.117480>.
- [90] Bradford, S.A., Schijven, J., 2002. Release of Cryptosporidium and Giardia from Dairy Calf Manure: Impact of Solution Salinity. *Environ Sci Technol* 36, 3916–3923. <https://doi.org/10.1021/es025573l>.
- [91] Bradford, S.A., Simunek, J., Bettahar, M., van Genuchten, M.T.H., Yates, S.R., 2003. Modeling Colloid Attachment, Straining, and Exclusion in Saturated Porous Media. *Environ Sci Technol* 37, 2242–2250. <https://doi.org/10.1021/es025899u>.
- [92] Guber, A.K., Karns, J.S., Pachepsky, Y.A., Sadeghi, A.M., Van Kessel, J.S., Dao, T. H., 2007. Comparison of release and transport of manure-borne Escherichia coli and enterococci under grass buffer conditions. *Lett Appl Microbiol* 44, 161–167. <https://doi.org/10.1111/j.1472-765X.2006.02065.x>.
- [93] Jong, M.-C., Harwood, C.R., Blackburn, A., Snape, J.R., Graham, D.W., 2020. Impact of redox conditions on antibiotic resistance conjugative gene transfer frequency and plasmid fate in wastewater ecosystems. *Environ Sci Technol* 54, 14984–14993. <https://doi.org/10.1021/acs.est.0c03714>.
- [94] Berglund, B., 2015. Environmental dissemination of antibiotic resistance genes and correlation to anthropogenic contamination with antibiotics. *Infect Ecol Epidemiol* 5, 28564. <https://doi.org/10.3402/iee.v5.28564>.
- [95] Zainab, S.M., Junaid, M., Xu, N., Malik, R.N., 2020. Antibiotics and antibiotic resistant genes (ARGs) in groundwater: a global review on dissemination, sources, interactions, environmental and human health risks. *Water Res* 187, 116455. <https://doi.org/10.1016/j.watres.2020.116455>.
- [96] Cho, K.H., Pachepsky, Y.A., Oliver, D.M., Muirhead, R.W., Park, Y., Quilliam, R. S., Shelton, D.R., 2016. Modeling fate and transport of fecally-derived microorganisms at the watershed scale: State of the science and future opportunities. *Water Res* 100, 38–56. <https://doi.org/10.1016/j.watres.2016.04.064>.
- [97] Brinkmeyer, R., Amon, R.M.W., Schwarz, J.R., Saxton, T., Roberts, D., Harrison, S., Ellis, N., Fox, J., DiGuardi, K., Hochman, M., Duan, S., Stein, R., Elliott, C., 2015. Distribution and persistence of Escherichia coli and Enterococci in stream bed and bank sediments from two urban streams in Houston, TX. *Sci Total Environ* 502, 650–658. <https://doi.org/10.1016/j.scitotenv.2014.09.071>.
- [98] Piorkowski, G., Jamieson, R., Bezanson, G., Truelstrup Hansen, L., Yost, C., 2014. Reach specificity in sediment E. coli population turnover and interaction with waterborne populations. *Sci Total Environ* 496, 402–413. <https://doi.org/10.1016/j.scitotenv.2014.06.145>.
- [99] Brown, P.C., Borowska, E., Schwartz, T., Horn, H., 2019. Impact of the particulate matter from wastewater discharge on the abundance of antibiotic resistance genes and facultative pathogenic bacteria in downstream river sediments. *Sci Total Environ* 649, 1171–1178. <https://doi.org/10.1016/j.scitotenv.2018.08.394>.
- [100] Chen, H., Jing, L., Yao, Z., Meng, F., Teng, Y., 2019. Prevalence, source and risk of antibiotic resistance genes in the sediments of Lake Tai (China) deciphered by metagenomic assembly: A comparison with other global lakes. *Environ Int* 127, 267–275. <https://doi.org/10.1016/j.envint.2019.03.048>.
- [101] Quintela-Baluja, M., Frigon, D., Abouelnaga, M., Jobling, K., Romalde, J.L., Gomez Lopez, M., Graham, D.W., 2021. Dynamics of integron structures across a wastewater network – Implications to resistance gene transfer. *Water Res* 206, 117720. <https://doi.org/10.1016/j.watres.2021.117720>.
- [102] Deng, C., Liu, X., Li, L., Shi, J., Guo, W., Xue, J., 2020. Temporal dynamics of antibiotic resistant genes and their association with the bacterial community in a water-sediment mesocosm under selection by 14 antibiotics. *Environ Int* 137, 105554. <https://doi.org/10.1016/j.envint.2020.105554>.
- [103] Duarte, D.J., Oldenkamp, R., Ragas, A.M.J., 2019. Modelling environmental antibiotic-resistance gene abundance: a meta-analysis. *Sci Total Environ* 659, 335–341. <https://doi.org/10.1016/j.scitotenv.2018.12.233>.
- [104] Vital, P.G., Zara, E.S., Paraoan, C.E.M., Dimasupil, M.A.Z., Abello, J.J.M., Santos, I.T.G., Rivera, W.L., 2018. Antibiotic resistance and extended-spectrum beta-lactamase production of escherichia coli isolated from irrigation waters in selected urban farms in Metro Manila, Philippines. *Water* 10, 548. <https://doi.org/10.3390/w10050548>.
- [105] Amos, G.C., Gozzard, E., Carter, C.E., Mead, A., Bowes, M.J., Hawkey, P.M., Zhang, L., Singer, A.C., Gaze, W.H., Wellington, E.M.H., 2015. Validated predictive modelling of the environmental resistome. *ISME J* 9, 1467–1476. <https://doi.org/10.1038/ismej.2014.237>.
- [106] Graham, D.W., Olivares-Rieumont, S., Knapp, C.W., Lima, L., Werner, D., Bowen, E., 2011. Antibiotic resistance gene abundances associated with waste discharges to the Almendares River near Havana, Cuba. *Environ Sci Technol* 45, 418–424. <https://doi.org/10.1021/es102473z>.
- [107] Hatosy, S.M., Martiny, A.C., 2015. The ocean as a global reservoir of antibiotic resistance genes. *Appl Environ Microbiol* 81, 7593–7599. <https://doi.org/10.1128/AEM.00736-15>.
- [108] Pruden, A., Arabi, M., Storteboom, H.N., 2012. Correlation between upstream human activities and riverine antibiotic resistance genes. *Environ Sci Technol* 46, 11541–11549. <https://doi.org/10.1021/es302657r>.
- [109] Czekalski, N., Gascón Díez, E., Bürgmann, H., 2014. Wastewater as a point source of antibiotic-resistance genes in the sediment of a freshwater lake. *ISME J* 8, 1381–1390. <https://doi.org/10.1038/ismej.2014.8>.
- [110] Gothwal, R., Shashidhar, T., 2017. Proliferation of ciprofloxacin resistant bacteria in polluted sediments of Musi River, India. *Soil Sediment Contam Int J* 26, 501–509. <https://doi.org/10.1080/15320383.2017.1355352>.
- [111] Popowska, M., Rzezycka, M., Miernik, A., Krawczyk-Balska, A., Walsh, F., Duffy, B., 2012. Influence of soil use on prevalence of tetracycline, streptomycin, and erythromycin resistance and associated resistance genes. *Antimicrob Agents Chemother* 56, 1434–1443. <https://doi.org/10.1128/AAC.05766-11>.
- [112] Sabri, N.A., Schmitt, H., Van der Zaaij, B., Gerritsen, H.W., Zuidema, T., Rijnaarts, H.H.M., Langenhoff, A.A.M., 2020. Prevalence of antibiotics and antibiotic resistance genes in a wastewater effluent-receiving river in the Netherlands. *J Environ Chem Eng* 8, 102245. <https://doi.org/10.1016/j.jece.2018.03.004>.
- [113] Knapp, C., Lima, L., Olivares-Rieumont, S., Bowen, E., Werner, D., Graham, D.W., 2012. Seasonal Variations in Antibiotic Resistance Gene Transport in the Almendares River, Havana, Cuba. *Front Microbiol* 3.
- [114] Ott, A., O'Donnell, G., Tran, N.H., Mohd Haniffah, M.R., Su, J.-Q., Zealand, A.M., Gin, K.Y.-H., Goodson, M.L., Zhu, Y.-G., Graham, D.W., 2021. Developing surrogate markers for predicting antibiotic resistance “Hot Spots” in rivers where limited data are available. *Environ Sci Technol* 55, 7466–7478. <https://doi.org/10.1021/acs.est.1c00939>.
- [115] Davis, B.C., Riquelme, M.V., Ramirez-Toro, G., Bandaragoda, C., Garner, E., Rhoads, W.J., Vikesland, P., Pruden, A., 2020. Demonstrating an integrated antibiotic resistance gene surveillance approach in Puerto Rican watersheds post-hurricane maria. *Environ Sci Technol* 54, 15108–15119. <https://doi.org/10.1021/acs.est.0c05567>.
- [116] Liang, S.Y., Messenger, N., 2018. Infectious diseases after hydrologic disasters. *Emerg Med Clin* 36, 835–851. <https://doi.org/10.1016/j.emc.2018.07.002>.
- [117] Cesare, A.D., Eckert, E.M., Rogora, M., Corno, G., 2017. Rainfall increases the abundance of antibiotic resistance genes within a riverine microbial community. *Environ Pollut* 226, 473–478. <https://doi.org/10.1016/j.envpol.2017.04.036>.
- [118] Ahmed, W., Zhang, Q., Lobos, A., Senkbeil, J., Sadowsky, M.J., Harwood, V.J., Saeidi, N., Marinoni, O., Ishii, S., 2018. Precipitation influences pathogenic bacteria and antibiotic resistance gene abundance in storm drain outfalls in coastal sub-tropical waters. *Environ Int* 116, 308–318. <https://doi.org/10.1016/j.envint.2018.04.005>.

- [119] Hayward, J.L., Jackson, A.J., Yost, C.K., Truelstrup Hansen, L., Jamieson, R.C., 2018. Fate of antibiotic resistance genes in two Arctic tundra wetlands impacted by municipal wastewater. *Sci Total Environ* 642, 1415–1428. <https://doi.org/10.1016/j.scitotenv.2018.06.083>.
- [120] Lee, J., Ju, F., Maile-Moskowitz, A., Beck, K., Maccagnan, A., McArdell, C.S., Dal Molin, M., Fenicia, F., Vikesland, P.J., Pruden, A., Stamm, C., Bürgmann, H., 2021. Unraveling the riverine antibiotic resistome: The downstream fate of anthropogenic inputs. *Water Res* 197, 117050. <https://doi.org/10.1016/j.watres.2021.117050>.
- [121] Rusconi, R., Guasto, J.S., Stocker, R., 2014. Bacterial transport suppressed by fluid shear. *Nat Phys* 10, 212–217. <https://doi.org/10.1038/nphys2883>.
- [122] Aslan, A., Cole, Z., Bhattacharya, A., Oyibo, O., 2018. Presence of antibiotic-resistant *Escherichia coli* in wastewater treatment plant effluents utilized as water reuse for irrigation. *Water* 10, 805. <https://doi.org/10.3390/w10060805>.
- [123] Chen, Y., Chen, H., Zhang, L., Jiang, Y., Gin, K.Y.-H., He, Y., 2018. Occurrence, distribution, and risk assessment of antibiotics in a subtropical river-reservoir system. *Water* 10, 104. <https://doi.org/10.3390/w10020104>.
- [124] Chandrasekar, A., Binder, M., Liedl, R., U Berendonk, T., 2021. Reactive-transport modelling of *Enterococcus faecalis* JH2-2 passage through water saturated sediment columns. *J Hazard Mater* 413, 125292. <https://doi.org/10.1016/j.jhazmat.2021.125292>.
- [125] Jeon, D.J., Ligaray, M., Kim, M., Kim, G., Lee, G., Pachepsky, Y.A., Cha, D.-H., Cho, K.H., 2019. Evaluating the influence of climate change on the fate and transport of fecal coliform bacteria using the modified SWAT model. *Sci Total Environ* 658, 753–762. <https://doi.org/10.1016/j.scitotenv.2018.12.213>.
- [126] Redshaw, C.H., Stahl-Timmins, W.M., Fleming, L.E., Davidson, I., Depledge, M.H., 2013. Potential changes in disease patterns and pharmaceutical use in response to climate change. *J Toxicol Environ Health Part B* 16, 285–320. <https://doi.org/10.1080/10937404.2013.802265>.
- [127] Burnham, J.P., 2021. Climate change and antibiotic resistance: a deadly combination. *Ther Adv Infect Dis* 8, 2049936121991374. <https://doi.org/10.1177/2049936121991374>.
- [128] Mohsin, M., Shad, A.A., Ali, J., Sajjad-ur-Rahman, 2020. Antimicrobial Resistance, Food Systems and Climate Change. In: Panwar, H., Sharma, C., Lichtfouse, E. (Eds.), *Sustainable Agriculture Reviews 46: Mitigation of Antimicrobial Resistance Vol 1 Tools and Targets, Sustainable Agriculture Reviews*. Springer International Publishing, Cham, pp. 59–81. https://doi.org/10.1007/978-3-030-53024-2_3.
- [129] Rodríguez-Verdugo, A., Lozano-Huntelman, N., Cruz-Loya, M., Savage, V., Yeh, P., 2020. Compounding effects of climate warming and antibiotic resistance. *iScience* 23, 101024. <https://doi.org/10.1016/j.isci.2020.101024>.
- [130] Eckert, E.M., Di Cesare, A., Coci, M., Corno, G., 2018. Persistence of antibiotic resistance genes in large subalpine lakes: the role of anthropogenic pollution and ecological interactions. *Hydrobiologia* 824, 93–108. <https://doi.org/10.1007/s10750-017-3480-0>.
- [131] Letten, A.D., Hall, A.R., Levine, J.M., 2021. Using ecological coexistence theory to understand antibiotic resistance and microbial competition. *Nat Ecol Evol* 5, 431–441. <https://doi.org/10.1038/s41559-020-01385-w>.
- [132] Fricker, C.R., 2003. The presence of bacteria in water after regrowth. in: *Heterotrophic Plate Counts and Drinking-Water Safety*. IWA Publish., London, pp. 49–60.
- [133] Wu, D., Zhao, J., Su, Y., Yang, M., Dolfing, J., Graham, D.W., Yang, K., Xie, B., 2023. Explaining the resistomes in a megacity's water supply catchment: Roles of microbial assembly-dominant taxa, niched environments and pathogenic bacteria. *Water Res* 228, 119359. <https://doi.org/10.1016/j.watres.2022.119359>.
- [134] Obermeier, M.M., Wicaksono, W.A., Taffner, J., Bergna, A., Poehlein, A., Cernava, T., Lindstaedt, S., Lovric, M., Müller Bogotá, C.A., Berg, G., 2021. Plant resistome profiling in evolutionary old bog vegetation provides new clues to understand emergence of multi-resistance. *ISME J* 15, 921–937. <https://doi.org/10.1038/s41396-020-00822-9>.
- [135] Grossart, H.-P., Dziallas, C., Leunert, F., Tang, K.W., 2010. Bacteria dispersal by hitchhiking on zooplankton. *Proc Natl Acad Sci* 107, 11959–11964. <https://doi.org/10.1073/pnas.1000668107>.
- [136] Eckert, E.M., Di Cesare, A., Stenzel, B., Fontaneto, D., Corno, G., 2016. *Daphnia* as a refuge for an antibiotic resistance gene in an experimental freshwater community. *Sci Total Environ* 571, 77–81. <https://doi.org/10.1016/j.scitotenv.2016.07.141>.
- [137] Colombo, S., Arioli, S., Neri, E., Della Scala, G., Gargari, G., Mora, D., 2017. Viromes as genetic reservoir for the microbial communities in aquatic environments: a focus on antimicrobial-resistance genes. *Front Microbiol* 8. <https://doi.org/10.3389/fmicb.2017.01095>.
- [138] González-Zorn, B., Escudero, J.A., 2012. Ecology of antimicrobial resistance: humans, animals, food and environment. *Int Microbiol* 101–109. <https://doi.org/10.2436/20.1501.01.163>.
- [139] Kraemer, S.A., Ramachandran, A., Perron, G.G., 2019. Antibiotic Pollution in the Environment: From Microbial Ecology to Public Policy. *Microorganisms* 7, 180. <https://doi.org/10.3390/microorganisms7060180>.
- [140] Greenfield, B.K., Shaked, S., Marrs, C.F., Nelson, P., Raxter, I., Xi, C., McKone, T.E., Jolliet, O., 2011. Modeling the Emergence of Antibiotic Resistance in the Environment: an Analytical Solution for the Minimum Selection Concentration. *Antimicrob Agents Chemother* 62, e01686-17. <https://doi.org/10.1128/AAC.01686-17>.
- [141] Huijbers, P.M.C., Flach, C.-F., Larsson, D.G.J., 2019. A conceptual framework for the environmental surveillance of antibiotics and antibiotic resistance. *Environ Int* 130, 104880. <https://doi.org/10.1016/j.envint.2019.05.074>.
- [142] Gullberg, E., Cao, S., Berg, O.G., Ilbäck, C., Sandegren, L., Hughes, D., Andersson, D.I., 2011. Selection of Resistant Bacteria at Very Low Antibiotic Concentrations. *PLOS Pathog* 7, e1002158. <https://doi.org/10.1371/journal.ppat.1002158>.
- [143] Yin, H., Chen, X., Li, G., Chen, Y., Wang, W., An, T., Wong, P.K., Zhao, H., 2019. Sub-lethal photocatalysis bactericidal technology cause longer persistence of antibiotic-resistance mutant and plasmid through the mechanism of reduced fitness cost. *Appl Catal B Environ* 245, 698–705. <https://doi.org/10.1016/j.apcatb.2019.01.041>.
- [144] Pachepsky, Y.A., Allende, A., Boithias, L., Cho, K., Jamieson, R., Hofstra, N., Molina, M., 2018. Microbial water quality: monitoring and modeling. *J Environ Qual* 47, 931–938. <https://doi.org/10.2134/jeq2018.07.0277>.
- [145] Hornberger, G.M., Mills, A.L., Herman, J.S., 1992. Bacterial transport in porous media: evaluation of a model using laboratory observations. *Water Resour Res* 28, 915–923. <https://doi.org/10.1029/91WR02980>.
- [146] Tanji, Y., Mizoguchi, K., Yoichi, M., Morita, M., Kijima, N., Kator, H., Unno, H., 2003. Seasonal change and fate of coliphages infected to *Escherichia coli* O157:H7 in a wastewater treatment plant. *Water Res* 37, 1136–1142. [https://doi.org/10.1016/S0043-1354\(02\)00464-5](https://doi.org/10.1016/S0043-1354(02)00464-5).
- [147] Blaustein, R.A., Pachepsky, Y., Hill, R.L., Shelton, D.R., Whelan, G., 2013. *Escherichia coli* survival in waters: Temperature dependence. *Water Res* 47, 569–578. <https://doi.org/10.1016/j.watres.2012.10.027>.
- [148] Corapcioglu, M.Y., Haridas, A., 1985. Microbial transport in soils and groundwater: a numerical model. *Adv Water Resour* 8, 188–200. [https://doi.org/10.1016/0309-1708\(85\)90063-6](https://doi.org/10.1016/0309-1708(85)90063-6).
- [149] Benham, B.L., Baffaut, C., Zeckoski, R.W., Mankin, K.R., Pachepsky, Y.A., Sadeghi, A.M., Brannan, K.M., Soupir, M.L., Habersack, M.J., 2006. Modeling Bacteria Fate and Transport in Watersheds to Support TMDLs. *Trans ASABE* 49, 987–1002. <https://doi.org/10.13031/2013.21739>.
- [150] Oliver, D.M., Porter, K.D.H., Pachepsky, Y.A., Muirhead, R.W., Reaney, S.M., Coffey, R., Kay, D., Milledge, D.G., Hong, E., Anthony, S.G., Page, T., Bloodworth, J.W., Mellander, P.-E., Carbonneau, P.E., McGrane, S.J., Quilliam, R. S., 2016. Predicting microbial water quality with models: Over-arching questions for managing risk in agricultural catchments. *Sci Total Environ* 544, 39–47. <https://doi.org/10.1016/j.scitotenv.2015.11.086>.
- [151] Bradford, S.A., Simunek, J., Walker, S.L., 2006. Transport and straining of *E. coli* O157:H7 in saturated porous media. *Water Resour Res* 42. <https://doi.org/10.1029/2005WR004805>.
- [152] Xu, S., Liao, Q., Saiers, J.E., 2008. Straining of nonspherical colloids in saturated porous media. *Environ Sci Technol* 42, 771–778. <https://doi.org/10.1021/es071328w>.
- [153] Harter, T., Wagner, S., Atwill, E.R., 2000. Colloid Transport and Filtration of *Cryptosporidium parvum* in Sandy Soils and Aquifer Sediments. *Environ Sci Technol* 34, 62–70. <https://doi.org/10.1021/es990132w>.
- [154] Pérez-Vidal, A., Rivera-Sánchez, S.P., Florez-Elvira, L.J., Silva-Leal, J.A., Diaz-Gomez, J., Herrera-Cuero, L.F., Lopez Botero, L.P., 2019. Removal of *E. coli* and *Salmonella* in pot ceramic filters operating at different filtration rates. *Water Res* 159, 358–364. <https://doi.org/10.1016/j.watres.2019.05.028>.
- [155] Sadeghi, G., Schijven, J.F., Behrends, T., Hassanizadeh, S.M., van Genuchten, M. Th., 2013. Bacteriophage PRD1 batch experiments to study attachment, detachment and inactivation processes. *J Contam Hydrol* 152, 12–17. <https://doi.org/10.1016/j.jconhyd.2013.06.002>.
- [156] Zheng, X., Shan, B., Chen, L., Sun, Y., Zhang, S., 2014. Attachment-detachment dynamics of suspended particle in porous media: experiment and modeling. *J Hydrol* 511, 199–204. <https://doi.org/10.1016/j.jhydrol.2014.01.039>.
- [157] Smith, E., Badawy, A., 2008. Modelling *E. coli* transport in soil columns: simulation of wastewater reuse in agriculture. *Water Sci Technol J Int Assoc Water Pollut Res* 57, 1123–1129. <https://doi.org/10.2166/wst.2008.205>.
- [158] Xu, S., Gao, B., Saiers, J.E., 2006. Straining of colloidal particles in saturated porous media. *Water Resour Res* 42. <https://doi.org/10.1029/2006WR004948>.
- [159] Bradford, S.A., Yates, S.R., Bettahar, M., Simunek, J., 2002. Physical factors affecting the transport and fate of colloids in saturated porous media. *Water Resour Res* 38 (12), 631–6363. <https://doi.org/10.1029/2002WR001340>.
- [160] Stumpp, C., Lawrence, J.R., Hendry, M.J., Maloszewski, P., 2011. Transport and bacterial interactions of three bacterial strains in saturated column experiments. *Environ Sci Technol* 45, 2116–2123. <https://doi.org/10.1021/es103569u>.
- [161] Iqbal, M.S., Hofstra, N., 2019. Modeling *Escherichia coli* fate and transport in the Kabul River Basin using SWAT. *Hum Ecol Risk Assess Int J* 25, 1279–1297. <https://doi.org/10.1080/10807039.2018.1487276>.
- [162] Niazi, M., Obropta, C., Miskewitz, R., 2015. Pathogen transport and fate modeling in the Upper Salem River Watershed using SWAT model. *J Environ Manag* 151, 167–177. <https://doi.org/10.1016/j.jenvman.2014.12.042>.
- [163] Worku Meshesha, T., Wang, J., Demelash Melaku, N., 2020. A modified hydrological model for assessing effect of pH on fate and transport of *Escherichia coli* in the Athabasca River basin. *J Hydrol* 582, 124513. <https://doi.org/10.1016/j.jhydrol.2019.124513>.
- [164] Cho, K.H., Pachepsky, Y.A., Kim, M., Pyo, J., Park, M.-H., Kim, Y.M., Kim, J.-W., Kim, J.H., 2016. Modeling seasonal variability of fecal coliform in natural surface waters using the modified SWAT. *J Hydrol* 535, 377–385. <https://doi.org/10.1016/j.jhydrol.2016.01.084>.
- [165] Yakirevich, A., Pachepsky, Y.A., Guber, A.K., Gish, T.J., Shelton, D.R., Cho, K.H., 2013. Modeling transport of *Escherichia coli* in a creek during and after artificial high-flow events: Three-year study and analysis. *Water Res* 47, 2676–2688. <https://doi.org/10.1016/j.watres.2013.02.011>.
- [166] Yates, M.V., Ouyang, Y., 1992. VIRTUS, a model of virus transport in unsaturated soils. *Appl Environ Microbiol* 58, 1609–1616.

- [167] Drummond, J.D., Davies-Colley, R.J., Stott, R., Sukias, J.P., Nagels, J.W., Sharp, A., Packman, A.I., 2015. Microbial transport, retention, and inactivation in streams: a combined experimental and stochastic modeling approach. *Environ Sci Technol* 49, 7825–7833. <https://doi.org/10.1021/acs.est.5b01414>.
- [168] Ferguson, C., Husman, A.M. de R., Altavilla, N., Deere, D., Ashbolt, N., 2003. Fate and Transport of Surface Water Pathogens in Watersheds. *Crit Rev Environ Sci Technol* 33, 299–361. <https://doi.org/10.1080/10643380390814497>.
- [169] Ng, C., Chen, H., Goh, S.G., Haller, L., Wu, Z., Charles, F.R., Trotter, A., Gin, K., 2018. Microbial water quality and the detection of multidrug resistant *E. coli* and antibiotic resistance genes in aquaculture sites of Singapore. *Mar Pollut Bull* 135, 475–480. <https://doi.org/10.1016/j.marpolbul.2018.07.055>.
- [170] Rodríguez-Mozaz, S., Chamorro, S., Martí, E., Huerta, B., Gros, M., Sánchez-Melió, A., Borrego, C.M., Barceló, D., Balcázar, J.L., 2015. Occurrence of antibiotics and antibiotic resistance genes in hospital and urban wastewaters and their impact on the receiving river. *Water Res* 69, 234–242. <https://doi.org/10.1016/j.watres.2014.11.021>.
- [171] Tran, N.H., Hoang, L., Nghiem, L.D., Nguyen, N.M.H., Ngo, H.H., Guo, W., Trinh, Q.T., Mai, N.H., Chen, H., Nguyen, D.D., Ta, T.T., Gin, K.Y.-H., 2019. Occurrence and risk assessment of multiple classes of antibiotics in urban canals and lakes in Hanoi, Vietnam. *Sci Total Environ* 692, 157–174. <https://doi.org/10.1016/j.scitotenv.2019.07.092>.
- [172] Niewiadomska, A.M., Jayabalasingham, B., Seidman, J.C., Willem, L., Grenfell, B., Spiro, D., Viboud, C., 2019. Population-level mathematical modeling of antimicrobial resistance: a systematic review. *BMC Med* 17, 81. <https://doi.org/10.1186/s12916-019-1314-9>.
- [173] Chandrasekaran, S., Jiang, S.C., 2019. A dose response model for quantifying the infection risk of antibiotic-resistant bacteria. *Sci Rep* 9, 17093. <https://doi.org/10.1038/s41598-019-52947-3>.
- [174] Spicknall, I.H., Foxman, B., Marrs, C.F., Eisenberg, J.N.S., 2013. A modeling framework for the evolution and spread of antibiotic resistance: literature review and model categorization. *Am J Epidemiol* 178, 508–520. <https://doi.org/10.1093/aje/kwt017>.
- [175] Trauer, J.M., Achar, J., Parpieva, N., Khamraev, A., Denholm, J.T., Falzon, D., Jaramillo, E., Mesic, A., du Cros, P., McBryde, E.S., 2016. Modelling the effect of short-course multidrug-resistant tuberculosis treatment in Karakalpakstan, Uzbekistan. *BMC Med* 14. <https://doi.org/10.1186/s12916-016-0723-2>.
- [176] Brauwere, A., de Ouattara, N.K., Servais, P., 2014. Modeling fecal indicator bacteria concentrations in natural surface waters: a review. *Crit Rev Environ Sci Technol* 44, 2380–2453. <https://doi.org/10.1080/10643389.2013.829978>.
- [177] Leclerc, Q.J., Lindsay, J.A., Knight, G.M., 2019. *Math Model Study Horiz Transf Antimicrob Resist Genes Bact: Curr State Field Recomm* 10.
- [178] Gothwal, R., Thatikonda, S., 2018. Mathematical model for the transport of fluoroquinolone and its resistant bacteria in aquatic environment. *Environ Sci Pollut Res* 25, 20439–20452. <https://doi.org/10.1007/s11356-017-9848-x>.
- [179] Gothwal, R., Thatikonda, S., 2020. Modeling transport of antibiotic resistant bacteria in aquatic environment using stochastic differential equations. *Sci Rep* 10, 15081. <https://doi.org/10.1038/s41598-020-72106-3>.
- [180] Hellweger, F.L., 2013. Simple model of tetracycline antibiotic resistance in aquatic environment: accounting for metal coselection. *J Environ Eng* 139, 913–921. [https://doi.org/10.1061/\(ASCE\)EE.1943-7870.0000696](https://doi.org/10.1061/(ASCE)EE.1943-7870.0000696).
- [181] Hellweger, F.L., Ruan, X., Sanchez, S., 2011. A simple model of tetracycline antibiotic resistance in the aquatic environment (with Application to the Poudre River). *Int J Environ Res Public Health* 8, 480–497. <https://doi.org/10.3390/ijerph8020480>.
- [182] Ikuma, K., Rehmann, C.R., 2020. Importance of Extracellular DNA in the Fate and Transport of Antibiotic Resistance Genes Downstream of a Wastewater Treatment Plant. *Environ Eng Sci* 37, 164–168. <https://doi.org/10.1089/ees.2019.0319>.
- [183] Stange, C., Tiehm, A., 2020. Occurrence of antibiotic resistance genes and microbial source tracking markers in the water of a karst spring in Germany. *Sci Total Environ* 742, 140529. <https://doi.org/10.1016/j.scitotenv.2020.140529>.
- [184] Fiorentino, A., De Luca, G., Rizzo, L., Viccione, G., Lofrano, G., Carotenuto, M., 2018. Simulating the fate of indigenous antibiotic resistant bacteria in a mild slope wastewater polluted stream. *J Environ Sci* 69, 95–104. <https://doi.org/10.1016/j.jes.2017.04.018>.
- [185] Gothwal, R., Thatikonda, S., 2021. Modeling fluoroquinolone resistance in polluted aquatic environment of a river. *J Hazard Toxic Radioact Waste* 25, 04020080. [https://doi.org/10.1061/\(ASCE\)HZ.2153-5515.0000591](https://doi.org/10.1061/(ASCE)HZ.2153-5515.0000591).
- [186] Heijnsbergen, E., van, Niebaum, G., Lämmchen, V., Borneman, A., Hernández Leal, L., Klasmeyer, J., Schmitt, H., 2022. Antibiotic-Resistant *E. coli* in the Dutch–German Vecht Catchment—Monitoring and Modeling. *Environ Sci Technol*. <https://doi.org/10.1021/acs.est.2c00218>.
- [187] Rico, A., Jacobs, R., Van den Brink, P.J., Tello, A., 2017. A probabilistic approach to assess antibiotic resistance development risks in environmental compartments and its application to an intensive aquaculture production scenario. *Environ Pollut* 231, 918–928. <https://doi.org/10.1016/j.envpol.2017.08.079>.
- [188] Matthews, S., Rao, V.S., Durvasula, R.V., 2011. Modeling horizontal gene transfer (HGT) in the gut of the Chagas disease vector *Rhodnius prolixus*. *Parasit Vectors* 4, 77. <https://doi.org/10.1186/1756-3305-4-77>.
- [189] Song, W., Steensen, K., Thomas, T., 2017. HgtSIM: a simulator for horizontal gene transfer (HGT) in microbial communities. *PeerJ* 5, e4015. <https://doi.org/10.7717/peerj.4015>.
- [190] Mao, J., Lu, T., 2016. Population-dynamic modeling of bacterial horizontal gene transfer by natural transformation. *Biophys J* 110, 258–268. <https://doi.org/10.1016/j.bpj.2015.11.033>.
- [191] Massoudieh, A., Crain, C., Lambertini, E., Nelson, K.E., Barkouki, T., L'Amoreaux, P., Loge, F.J., Ginn, T.R., 2010. Kinetics of conjugative gene transfer on surfaces in granular porous media. *J Contam Hydrol, Front React Transp: Microb Dyn nad Redox Zonation Subsurf* 112, 91–102. <https://doi.org/10.1016/j.jconhyd.2009.10.009>.
- [192] Nielsen, K.M., Townsend, J.P., 2004. Monitoring and modeling horizontal gene transfer. *Nat Biotechnol* 22, 1110–1114. <https://doi.org/10.1038/nbt1006>.
- [193] Hunter, P.R., Wilkinson, D.C., Catling, L.A., Barker, G.C., 2008. Meta-analysis of experimental data concerning antimicrobial resistance gene transfer rates during conjugation. *Appl Environ Microbiol* 74, 6085–6090. <https://doi.org/10.1128/AEM.01036-08>.
- [194] Levin, B.R., Stewart, F.M., Rice, V.A., 1979. The kinetics of conjugative plasmid transmission: Fit of a simple mass action model. *Plasmid* 2, 247–260. [https://doi.org/10.1016/0147-619X\(79\)90043-X](https://doi.org/10.1016/0147-619X(79)90043-X).
- [195] Simonsen, L., Gordon, D.M., Stewart, F.M., Levin, B.R., 1990. Estimating the rate of plasmid transfer: an end-point method. *J Gen Microbiol* 136, 2319–2325. <https://doi.org/10.1099/00221287-136-11-2319>.
- [196] Levin, B.R., Udekwi, K.I., 2010. Population Dynamics of Antibiotic Treatment: a Mathematical Model and Hypotheses for Time-Kill and Continuous-Culture Experiments. *Antimicrob Agents Chemother* 54, 3414–3426. <https://doi.org/10.1128/AAC.00381-10>.
- [197] Knight, G.M., Costelloe, C., Deeny, S.R., Moore, L.S.P., Hopkins, S., Johnson, A.P., Robotham, J.V., Holmes, A.H., 2018. Quantifying where human acquisition of antibiotic resistance occurs: a mathematical modelling study. *BMC Med* 16, 137. <https://doi.org/10.1186/s12916-018-1121-8>.
- [198] Allen, H.K., Donato, J., Wang, H.H., Cloud-Hansen, K.A., Davies, J., Handelsman, J., 2010. Call of the wild: antibiotic resistance genes in natural environments. *Nat Rev Microbiol* 8, 251–259. <https://doi.org/10.1038/nrmicro2312>.
- [199] Andersson, D.I., Hughes, D., 2010. Antibiotic resistance and its cost: is it possible to reverse resistance? *Nat Rev Microbiol* 8, 260–271. <https://doi.org/10.1038/nrmicro2319>.
- [200] Knapp, C.W., Engemann, C.A., Hanson, M.L., Keen, P.L., Hall, K.J., Graham, D.W., 2008. Indirect Evidence of Transposon-Mediated Selection of Antibiotic Resistance Genes in Aquatic Systems at Low-Level Oxytetracycline Exposures. *Environ Sci Technol* 42, 5348–5353. <https://doi.org/10.1021/es703199g>.
- [201] Chen, Y., Su, J.-Q., Zhang, J., Li, P., Chen, H., Zhang, B., Gin, K.Y.-H., He, Y., 2019. High-throughput profiling of antibiotic resistance gene dynamic in a drinking water river-reservoir system. *Water Res* 149, 179–189. <https://doi.org/10.1016/j.watres.2018.11.007>.
- [202] Kim, M., Boithias, L., Cho, K.H., Silvera, N., Thammahacksa, C., Latschack, K., Rochelle-Newall, E., Sengtaheuanghoung, O., Pierret, A., Pachepsky, Y.A., Ribolzi, O., 2017. Hydrological modeling of Fecal Indicator Bacteria in a tropical mountain catchment. *Water Res* 119, 102–113. <https://doi.org/10.1016/j.watres.2017.04.038>.
- [203] Sadeghi, A.M., Arnold, J.G., 2002. A SWAT/microbial sub-model for predicting pathogen loadings in surface and groundwater at watershed and basin scales, in: *Total Maximum Daily Load (TMDL): Environmental Regulations. Proc 2002 Conf Am Soc Agric Biol Eng* 56.
- [204] Bergstrom, C.T., Lo, M., Lipsitch, M., 2004. Ecological theory suggests that antimicrobial cycling will not reduce antimicrobial resistance in hospitals. *Proc Natl Acad Sci* 101, 13285–13290. <https://doi.org/10.1073/pnas.0402298101>.
- [205] Kleczkowski, A., Hoyle, A., McMenemy, P., 2019. One model to rule them all? Modelling approaches across OneHealth for human, animal and plant epidemics. *Philos Trans R Soc B Biol Sci* 374, 20180255. <https://doi.org/10.1098/rstb.2018.0255>.